CIS I Assignment 2

Kinematic Calibration of a Cartesian Robot

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Academic-integrity statement

I (we) have followed the rules in completing this assignment.

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Contents

1 Introduction

2 Variables and Notation
- 2.1 Conventions
- 2.2 Symbols
- 2.3 Exact quantities
- 2.4 Error terms and limits

3 Mathematical Approach
- 3.1 Goal
- 3.2 Observations and calibration parameters
- 3.3 Calibration pipeline
- 3.4 Error analysis and parameter uncertainty

4 Algorithmic Approach
- 4.1 Scene design (Step 1)
- 4.2 Homing (Step 6)
- 4.3 Pose planning (Step 7)
- 4.4 Data collection and nominal base estimate (Steps 6–7)
- 4.5 Decoupled initial estimate (Step 8)
- 4.6 Weighted least-squares fit (Step 8)
- 4.7 Rotation about $\boldsymbol{d}_1$ (Step 8)
- 4.8 Improved forward and inverse kinematics
- 4.9 Evaluation (Steps 9–10)
- 4.10 Function-level execution flow

5 Program Structure
- 5.1 File and module organization
- 5.2 Program hierarchy
- 5.3 Runtime functions
- 5.4 Tests
- 5.5 Program outputs

6 Validation Approach
- 6.1 End-to-end accuracy
- 6.2 Unit tests
- 6.3 Controlled sensitivity experiments
- 6.4 Limitations

7 Results
- 7.1 Calibration data and time
- 7.2 Estimated parameters
- 7.3 Accuracy
- 7.4 Several seeds and the two plans
- 7.5 Handout's base-frame definition
- 7.6 Additional function tests and controlled experiments

1 Introduction

2 Variables and Notation

2.1 Conventions

- Bold lowercase symbols are 3-vectors; lengths in mm, angles in rad. $F = \mathrm{Frame}(R, \boldsymbol{p})$, $F\boldsymbol{x} = R\boldsymbol{x} + \boldsymbol{p}$, $\mathrm{sk}(\boldsymbol{u})\boldsymbol{v} = \boldsymbol{u} \times \boldsymbol{v}$; small errors $\Delta F = \mathrm{Frame}(I + \mathrm{sk}(\boldsymbol{\alpha}), \boldsymbol{\epsilon})$.
- The rotation R and translation $\boldsymbol{p}$ of a pose F carry exactly the same subscripts, superscripts and prediction marks as F. For example, $F_{m,\mathrm{obs},j} = \mathrm{Frame}(R_{m,\mathrm{obs},j}, \boldsymbol{p}_{m,\mathrm{obs},j})$.
- Unmarked: nominal. *: actual (hidden). Subscript obs: observation; est: estimated parameter. A hat denotes a pose predicted by the kinematic model for given joint values. Poses with one subscript keep the handout's meaning: F_{tracker}, F_{rob} relative to the workspace, F_m relative to the tracker, F_{ee} relative to the robot base.
- F_X is the robot base relative to the tracker, the name used in the repository (`workflow.estimate_base_frame`); the handout gives this frame no symbol. The repository's `F_X_nom` is not the nominal F_X: it is an estimate from the observations computed with the nominal kinematics.
- j indexes calibration samples (as `data.thetas[j]`), k joints or markers. Superscripts rob and trk separate the robot's errors from the tracker's, as in the handout.
- *mBee* coincides with the end-effector frame, so its pose relative to the base is F_{ee} (the handout's F_{mBee}). The tool-mounting error $\Delta F_{\mathrm{ee}}(\boldsymbol{\eta}_{\mathrm{ee}})$ of the handout is written ΔF_{we}, because ΔF_{ee} also names the end-effector error in the accuracy definition.

2.2 Symbols

Table 1: Symbols and repository variables (ground truth for validation only).

Symbol	Meaning	Repository variable
Frames		
F_{tracker}	Tracker pose relative to the workspace; exact.	`F_trk`
F_{rob}, F^*_{rob}	Nominal and actual robot base pose relative to the workspace.	`rob.F_rob`; `ground_truth_parameters()["F_rob_actual"]`
F_X, F^*_X, $F_{X,\mathrm{est}}$	Nominal, actual and estimated base pose relative to the tracker, $F_X = F^{-1}_{\mathrm{tracker}} F_{\mathrm{rob}}$, $F^*_X = F^{-1}_{\mathrm{tracker}} F^*_{\mathrm{rob}}$. $F_{X,\mathrm{est}}$ denotes every estimate of it; the text states which.	`F_trk.inv()*rob.F_rob`; `F_X_baseline`, `F_X_nom`, `improved.base_frame()`

Symbol	Meaning	Repository variable
$F_{m,\mathrm{obs},j}$	Pose of *mBee* reported by the tracker in sample j: SVD registration of the reported markers, an estimate of $F_X^* F_{\mathrm{ee}}^*(\boldsymbol{\theta}_j)$ (the handout's F_m = MostRecentObservedFrame).	`data.frames[j]`
Robot chain		
$F_{\mathrm{ee}}(\boldsymbol{\theta})$	Nominal end-effector pose relative to the base, FwdKins($\boldsymbol{\theta}$).	`rob.fwd_kins(theta)`
$F_{\mathrm{ee}}^*(\boldsymbol{\theta})$	Actual end-effector pose relative to the actual base when the encoders read $\boldsymbol{\theta}$.	`rob.actual_ee_pose(theta)` (in the workspace)
$\hat{F}_{\mathrm{ee}}(\boldsymbol{\theta})$	Predicted calibrated pose, ImprovedFwdKins($\boldsymbol{\theta}$); its inverse is ImprovedInvKins.	`improved.fwd_kins`, `improved.inv_kins`
F_c, F_{cw}, F_w, F_{we}	Links of the chain: stage translation, Frame$(I, [500, 0, 0]^T)$, wrist rotation, Frame$(I, [100, 0, 0]^T)$.	Inside `rob.fwd_kins`; `Robot.F_CW`, `Robot.F_WE`
$\boldsymbol{d}_k$, $\boldsymbol{d}_k^*$, $k = 1, \ldots, 6$	Nominal and actual axis directions; stages for $k \leq 3$, wrist for $k \geq 4$ (the handout also writes the wrist axes as $\boldsymbol{n}_{k-3} = \boldsymbol{d}_k$).	`Robot.AXES`; `ground_truth_parameters()["axes"]`
$\boldsymbol{\phi}_{k,\mathrm{est}}$, $\Delta\theta_{k,\mathrm{sys,est}}^{\mathrm{rob}}$, $\Delta F_{\mathrm{we,est}}$	Estimated axis tilts, joint distortion and tool error.	`improved.params`, `improved.parameters_report()`
Joints, markers and time		
$\boldsymbol{\theta}$, $\boldsymbol{\theta}^*$	Encoder values (equal to the last command) and physical joint values.	`rob.state().theta`; `rob._theta_star`
$\boldsymbol{\theta}_j$	Commanded joint values of sample j.	`data.thetas[j]`
$\boldsymbol{\theta}_{\mathrm{encoders}}$	Encoder values returned by HomeRobot.	Return of `rob.home_robot()`
$\theta_k^{\min}$, $\theta_k^{\max}$, $\dot{\theta}_k^{\max}$, $\ddot{\theta}_k^{\max}$	Joint range, speed and acceleration limits.	`rob.theta_min`, ..., `rob.theta_ddot_max`
$\boldsymbol{a}_k^*$	Marker positions of *mBee* in the end-effector frame; exact.	`mbee_local_positions`
$\boldsymbol{b}_{k,\mathrm{obs},j}$, $\boldsymbol{b}_{k,j}^*$	Reported and actual tracker-frame position of *mBee* marker k in sample j.	`obs["mBee"].markers[k]`
t_{elap}, Δt_{samp}, t_{settle}	Simulated elapsed time, time per tracker observation, settling time per move.	`clock.elapsed`, `SAMPLE_DELTA_T_SEC`, `T_SETTLE`
Accuracy		
ΔF_{FwdK}; $\boldsymbol{\alpha}_{\mathrm{FwdK}}$, $\boldsymbol{\epsilon}_{\mathrm{FwdK}}$	Forward error and its rotation and translation.	`metrics.frame_error`
ΔF_{InvK}; $\boldsymbol{\alpha}_{\mathrm{InvK}}$, $\boldsymbol{\epsilon}_{\mathrm{InvK}}$	Inverse error and its rotation and translation.	`metrics.frame_error`

2.3 Exact quantities

Table 2: Quantities treated as exact.

Quantity	Handout basis	Code basis
Tracker pose	No jiggle, $\sigma_{Ta} = \sigma_{Tp} = 0$ (p. 3).	`TRACKER_JIGGLE_ANGLE_STD` = `TRACKER_JIGGLE_POS_STD` = 0.
F_{cw}	No error listed (p. 3).	`_actual_fwd_kins` uses the nominal `self.F_CW`.
First axis, $\boldsymbol{d}_1^* = \boldsymbol{d}_1$	Stated (p. 3).	`axes_actual[0]` is nominal; first tilt row is zero.
Orientation after the stages	F_c is a pure translation (p. 3), so F_{cw} always acts along the base x axis.	`Frame(Rot(),p_c)` in `_actual_fwd_kins`.
Wrist center	F_w is a product of rotations (p. 3); the wrist axes meet at one point.	Actual wrist rotations `exp_so3` carry no translation.
mBee markers and frame	Given $\boldsymbol{a}_k^*$ are actual; its frame is the end-effector frame (p. 5).	Zero manufacturing error (`simulation/calibration_body.py`).
Home position $\boldsymbol{\theta}^* = \mathbf{0}$	Stated (p. 4).	`home_robot` sets `_theta_star` to zero.
Encoder observation $\boldsymbol{\theta}$	State() returns θ_k (p. 3).	`state().theta` returns `theta_cmd` exactly.

2.4 Error terms and limits

Table 3: Error terms, models and limits. Static errors are drawn with norm between 0.5 and 1 times the bound.

Error term	Model	Bound or distribution	Drawn	Code
ΔF_{rob}; $\boldsymbol{\alpha}_{\text{rob}}$, $\boldsymbol{\epsilon}_{\text{rob}}$	$F_{\text{rob}}^* = \Delta F_{\text{rob}} F_{\text{rob}}$, $\Delta F_{\text{rob}} = \text{Frame}(\text{Rot}(\boldsymbol{\alpha}_{\text{rob}}), \boldsymbol{\epsilon}_{\text{rob}})$	$\lVert\boldsymbol{\alpha}_{\text{rob}}\rVert_2 \le \alpha_{\text{rob}}^{\max} = 0.05$; $\lVert\boldsymbol{\epsilon}_{\text{rob}}\rVert_2 \le \epsilon_{\text{rob}}^{\max} = 5\,\text{mm}$	Once per seed	`ALPHA_MAX_ROB`, `EPS_MAX_ROB`
$\Delta\theta_{k,\text{rnd}}^{\text{rob}}$	$\theta_k^* = \theta_k + \Delta\theta_{k,\text{rnd}}^{\text{rob}} + \Delta\theta_{k,\text{sys}}^{\text{rob}}$	$\mathcal{N}(0, \sigma_{k,\text{rnd}}^2)$, $\sigma_{k,\text{rnd}} = 0.005\,\text{mm}$ / 10^{-4}	Every move and homing	`SIGMA_RND`
$\Delta\theta_{k,\text{sys}}^{\text{rob}}$	Same equation; a function of θ_k	$\lvert\cdot\rvert \le \delta_{k,\text{sys}}^{\text{rob}} = 0.2\,\text{mm}$ / 0.01; $\lvert d\cdot/d\theta_k\rvert \le \omega_{k,\text{sys}}^{\text{rob}} = 5\times10^{-4}$ / 5×10^{-5}	Once per seed; smooth	`DELTA_SYS`, `OMEGA_SYS`
$\boldsymbol{\phi}_k$, $2 \le k \le 6$	$\boldsymbol{d}_k^* = R(\boldsymbol{\phi}_k)\boldsymbol{d}_k$, $\boldsymbol{\phi}_k \perp \boldsymbol{d}_k$	$\lVert\boldsymbol{\phi}_k\rVert_2 \le \phi_{\max} = 0.02$	Once per seed	`PHI_MAX`
ΔF_{we}; $\boldsymbol{\alpha}_{\text{we}}$, $\boldsymbol{\epsilon}_{\text{we}}$	$F_{\text{we}}^* = F_{\text{we}}\Delta F_{\text{we}}$	$\lVert\boldsymbol{\alpha}_{\text{we}}\rVert_2 \le 0.01$; $\lVert\boldsymbol{\epsilon}_{\text{we}}\rVert_2 \le 0.5\,\text{mm}$	Once per seed	`WE_ALPHA_MAX`, `WE_EPS_MAX`
$\boldsymbol{\epsilon}_b$	$\boldsymbol{b}^* = \boldsymbol{b}_{\text{obs}} + \boldsymbol{\epsilon}_{\text{sys}}^{\text{trk}}(\boldsymbol{b}_{\text{obs}}) + \boldsymbol{\epsilon}_b$	$\mathcal{N}(\mathbf{0}, \sigma_b^2 I)$, $\sigma_b = 0.1\,\text{mm}$	Every marker, every observation	`SENSOR_NOISE_STD`
$\boldsymbol{\epsilon}_{\text{sys}}^{\text{trk}}(\boldsymbol{b}_{\text{obs}})$	Same equation	$\lVert\cdot\rVert_2 \le \delta_{\text{sys}}^{\text{trk}} = 0.1\,\text{mm}$; $\lVert\nabla\cdot\rVert_2 \le \omega_{\text{sys}}^{\text{trk}} = 7.5\times10^{-4}$	Once per tracker; smooth	`TRACKER_SYS_MAX_MM`, `TRACKER_SYS_GRAD_MAX`

Error term	Model	Bound or distribution	Drawn	Code
Accuracy targets	ΔF_{FwdK}, ΔF_{InvK} over the work volume	$\|\boldsymbol{\alpha}\|_2 \le 10^{-3}$; $\|\boldsymbol{\epsilon}\|_2 \le 0.2\,\mathrm{mm}$	–	`TARGET_ANG_ERR_RAD`, `TARGET_POS_ERR_MM`

3 Mathematical Approach

We follow Method 1 of the handout: estimate the robot's kinematic parameters and use them to improve the forward and inverse kinematics. After stating the accuracy goal, we relate the observations to the unknown model parameters and derive the calibration from single-joint motions. The pipeline includes the joint pose fit and the improved forward and inverse kinematics; the following error analysis separates exact cancellations, initialization approximations and physical uncertainties. Notation follows the Variables and Notation section.

3.1 Goal

Table 4 states what must be achieved. The forward error compares the model with the actual pose for the same encoder values; the inverse error compares a target with the pose the robot actually reaches.

Table 4: Deliverables and accuracy targets (over the whole work volume).

Deliverable	Error measure	Target
$\hat{F}_{\mathrm{ee}}(\boldsymbol{\theta}) = \mathrm{ImprovedFwdKins}(\boldsymbol{\theta})$	$\Delta F_{\mathrm{FwdK}} = F^*_{\mathrm{ee}}(\boldsymbol{\theta})^{-1}\hat{F}_{\mathrm{ee}}(\boldsymbol{\theta})$	$\|\boldsymbol{\alpha}\|_2 \le 10^{-3}\,\mathrm{rad}$, $\|\boldsymbol{\epsilon}\|_2 \le 0.2\,\mathrm{mm}$
$\boldsymbol{\theta} = \mathrm{ImprovedInvKins}(F)$	$\Delta F_{\mathrm{InvK}} = F^{-1}F^*_{\mathrm{ee}}(\boldsymbol{\theta})$	same
Calibration procedure	simulated time t_{elap}	as short as possible

3.2 Observations and calibration parameters

To calibrate the kinematics, we first relate the observed poses $F_{m,\mathrm{obs},j}$ to the commanded joint values $\boldsymbol{\theta}_j$ and identify the unknown quantities. We then estimate these quantities from single-joint motions before refining them jointly.

3.2.1 From marker readings to an observed pose

Sample j is one move to the encoder values $\boldsymbol{\theta}_j$ followed by one tracker observation. *mBee* is fixed to the end-effector and its frame is the end-effector frame, so its actual pose relative to the tracker is

$$F^{-1}_{\mathrm{tracker}} F^*_{\mathrm{rob}} F^*_{\mathrm{ee}}(\boldsymbol{\theta}_j) = F^*_X\, F^*_{\mathrm{ee}}(\boldsymbol{\theta}_j), \tag{1}$$

where $F^*_{\mathrm{ee}}(\boldsymbol{\theta}_j)$ includes the random joint error $\Delta\theta^{\mathrm{rob}}_{k,\mathrm{rnd}}$ drawn for move j. The actual tracker-frame positions $\boldsymbol{b}^*_{k,j}$ of the *mBee* markers $\boldsymbol{a}^*_k$ are

$$\boldsymbol{b}^*_{k,j} = F^*_X\, F^*_{\mathrm{ee}}(\boldsymbol{\theta}_j)\, \boldsymbol{a}^*_k. \tag{2}$$

The reported positions $\boldsymbol{b}_{k,\mathrm{obs},j}$ are related to the actual positions $\boldsymbol{b}^*_{k,j}$ by (handout p. 2)

$$\boldsymbol{b}^*_{k,j} = \boldsymbol{b}_{k,\mathrm{obs},j} + \boldsymbol{\epsilon}^{\mathrm{trk}}_{\mathrm{sys}}(\boldsymbol{b}_{k,\mathrm{obs},j}) + \boldsymbol{\epsilon}_{b,k,j}, \tag{3}$$

where $\boldsymbol{\epsilon}_{b,k,j}$ is new in every observation and $\boldsymbol{\epsilon}^{\text{trk}}_{\text{sys}}$ is a fixed function of position. SampleMarkerBodies then returns the observed pose $F_{m,\text{obs},j}$ that best maps the known marker positions $\boldsymbol{a}^*_k$ onto the reported positions $\boldsymbol{b}_{k,\text{obs},j}$ (the handout's F_m with $m = mBee$):

$$F_{m,\text{obs},j} = \arg\min_F \sum_k \|F\boldsymbol{a}^*_k - \boldsymbol{b}_{k,\text{obs},j}\|_2^2 . \tag{4}$$

Because of the tracker errors $\boldsymbol{\epsilon}^{\text{trk}}_{\text{sys}}$ and $\boldsymbol{\epsilon}_{b,k,j}$, the actual tracker-frame pose $F^*_X F^*_{\text{ee}}(\boldsymbol{\theta}_j)$ differs from the observed pose $F_{m,\text{obs},j}$ by a small error frame $\Delta F_{m,j}$:

$$F^*_X\, F^*_{\text{ee}}(\boldsymbol{\theta}_j) = F_{m,\text{obs},j}\, \Delta F_{m,j}, \qquad \Delta F_{m,j} = \text{Frame}\big(I + \text{sk}(\boldsymbol{\alpha}_{m,j}), \boldsymbol{\epsilon}_{m,j}\big), \tag{5}$$

with $\Delta F_{m,j} = I$ when $\boldsymbol{\epsilon}^{\text{trk}}_{\text{sys}} = \boldsymbol{\epsilon}_{b,k,j} = \mathbf{0}$. The observation error $\Delta F_{m,j}$ is unknown. The observed pose $F_{m,\text{obs},j}$ therefore depends on both the base transform F^*_X and the end-effector pose $F^*_{\text{ee}}(\boldsymbol{\theta}_j)$. Predicting $F_{m,\text{obs},j}$ requires a model of their product $F^*_X F^*_{\text{ee}}(\boldsymbol{\theta}_j)$.

3.2.2 From joint values to the actual pose

We next express the end-effector pose $F^*_{\text{ee}}(\boldsymbol{\theta})$ relative to the actual robot base in terms of the physical joint values θ^*_k, axis directions $\boldsymbol{d}^*_k$ and tool error ΔF_{we}. For encoder values $\boldsymbol{\theta}$ the physical joint values are $\theta^*_k = \theta_k + \Delta\theta^{\text{rob}}_{k,\text{rnd}} + \Delta\theta^{\text{rob}}_{k,\text{sys}}(\theta_k)$, and the actual end-effector pose relative to the actual base is

$$F^*_{\text{ee}}(\boldsymbol{\theta}) = F^*_c\, F_{\text{cw}}\, F^*_w\, F_{\text{we}} \Delta F_{\text{we}}, \tag{6}$$

$$F^*_c = \text{Frame}\Big(I,\ \textstyle\sum_{k=1}^{3} \theta^*_k \boldsymbol{d}^*_k\Big), \qquad F^*_w = \text{Frame}\Big(\textstyle\prod_{k=4}^{6} R(\boldsymbol{d}^*_k, \theta^*_k),\ \mathbf{0}\Big),$$

with $\boldsymbol{d}^*_k = R(\boldsymbol{\phi}_k)\boldsymbol{d}_k$. The wrist rotations $R(\boldsymbol{d}^*_k, \theta^*_k)$ are applied in sequence, so $\boldsymbol{d}^*_5$ and $\boldsymbol{d}^*_6$ are expressed in the frames left by the preceding wrist rotations. The nominal model $F_{\text{ee}}(\boldsymbol{\theta})$ uses the same chain with $\theta^*_k \to \theta_k$, $\boldsymbol{d}^*_k \to \boldsymbol{d}_k$, $\Delta F_{\text{we}} \to I$. Each move draws a new random joint error $\Delta\theta^{\text{rob}}_{k,\text{rnd}}$, which remains fixed during subsequent tracker readings until the next move or homing operation.

3.2.3 Quantities to be estimated

Predicting $F^*_X F^*_{\text{ee}}(\boldsymbol{\theta}_j)$ requires the base transform F^*_X, axis tilts $\boldsymbol{\phi}_k$, tool error ΔF_{we} and six joint-distortion functions $\Delta\theta^{\text{rob}}_{k,\text{sys}}$. Table 5 summarizes these unknown quantities and their bounds. The known reference quantities are F_{tracker}, F_{cw}, $\boldsymbol{d}_1$, $\boldsymbol{a}^*_k$, and the physical home position $\boldsymbol{\theta}^* = \mathbf{0}$ reached by HomeRobot. The encoder values returned there, $\boldsymbol{\theta}_{\text{encoders}}$, are in general not zero: $\boldsymbol{\theta}_{\text{encoders}}$ contains the systematic joint errors $\Delta\theta^{\text{rob}}_{k,\text{sys}}$ and random joint errors $\Delta\theta^{\text{rob}}_{k,\text{rnd}}$ at home.

Table 5: Unknown quantities and their bounds.

Unknown	Count	Bound
Base F^*_X	6	0.05 rad, 5 mm
Stage tilts $\boldsymbol{\phi}_2, \boldsymbol{\phi}_3$	2×2	$\phi_{\max} = 0.02$
Wrist tilts $\boldsymbol{\phi}_4, \boldsymbol{\phi}_5, \boldsymbol{\phi}_6$	3×2	$\phi_{\max} = 0.02$
Tool ΔF_{we} ($\boldsymbol{\alpha}_{\text{we}}$, $\boldsymbol{\epsilon}_{\text{we}}$)	6	0.01 rad, 0.5 mm
Joint distortion $\Delta\theta^{\text{rob}}_{k,\text{sys}}(\cdot)$	6 functions	$\delta^{\text{rob}}_{k,\text{sys}}$, $\omega^{\text{rob}}_{k,\text{sys}}$ (value and slope)

A tilt has two components because a component of $\boldsymbol{\phi}_k$ along $\boldsymbol{d}_k$ does not move the axis. The joint distortions $\Delta\theta^{\text{rob}}_{k,\text{sys}}$ are functions rather than a finite set of unknown numbers. Before $\Delta\theta^{\text{rob}}_{k,\text{sys}}$ can enter a numerical least-squares problem, we must choose a finite representation consistent with the handout's value and slope bounds.

3.2.4 Distortion values at selected joint positions

We choose a piecewise-linear representation of each joint-distortion function $\Delta\theta^{\mathrm{rob}}_{k,\mathrm{sys}}$. For joint k, prescribe encoder positions $\theta_k^{(0)} < \cdots < \theta_k^{(L_k)}$ spanning the joint range, where the parenthesized superscript (l) indexes the selected positions, $l = 0, \ldots, L_k$; it is not a power. By contrast, $\theta_{k,j}$ denotes joint k in observation j. The positions $\theta_k^{(l)}$ are fixed; the distortion values $\Delta\theta^{\mathrm{rob}}_{k,\mathrm{sys}}(\theta_k^{(l)})$ are unknown calibration parameters. After fitting, their estimates are $\Delta\theta^{\mathrm{rob}}_{k,\mathrm{sys,est}}(\theta_k^{(l)})$.

For an input joint value θ_k between adjacent selected positions $\theta_k^{(l)}$ and $\theta_k^{(l+1)}$, define the interpolation weight

$$\lambda = \frac{\theta_k - \theta_k^{(l)}}{\theta_k^{(l+1)} - \theta_k^{(l)}}, \qquad \theta_k^{(l)} \le \theta_k \le \theta_k^{(l+1)}.$$

Here θ_k is the input at which the distortion is required, whereas $\theta_k^{(l)}$ and $\theta_k^{(l+1)}$ are fixed selected positions. The interpolated distortion is

$$\begin{aligned}\Delta\theta^{\mathrm{rob}}_{k,\mathrm{sys,est}}(\theta_k) = (1-\lambda)\,\Delta\theta^{\mathrm{rob}}_{k,\mathrm{sys,est}}(\theta_k^{(l)}) \\ + \lambda\,\Delta\theta^{\mathrm{rob}}_{k,\mathrm{sys,est}}(\theta_k^{(l+1)}).\end{aligned} \tag{7}$$

With exact distortion values at the selected positions, the handout's slope bound limits the interpolation error between $\theta_k^{(l)}$ and $\theta_k^{(l+1)}$ to $\frac{1}{2}\omega^{\mathrm{rob}}_{k,\mathrm{sys}}(\theta_k^{(l+1)} - \theta_k^{(l)})$. This is a conservative bound that does not use the difference between the endpoint distortion values; known endpoint values can yield a tighter bound. Errors in the estimated endpoint values $\Delta\theta^{\mathrm{rob}}_{k,\mathrm{sys,est}}(\theta_k^{(l)})$ contribute additional error.

3.2.5 Parameters

With each joint-distortion function represented by its values $\Delta\theta^{\mathrm{rob}}_{k,\mathrm{sys}}(\theta_k^{(l)})$ at the selected positions, all unknown quantities can now be collected in one finite parameter vector

$$\begin{aligned}\boldsymbol{x} = \big[&\boldsymbol{\alpha}_X;\ \boldsymbol{\epsilon}_X;\ t_{2,1}, \ldots, t_{6,2};\ \boldsymbol{\alpha}_{\mathrm{we}};\ \boldsymbol{\epsilon}_{\mathrm{we}};\\ &\big(\Delta\theta^{\mathrm{rob}}_{k,\mathrm{sys}}(\theta_k^{(l)})\big)_{k,l}\big].\end{aligned} \tag{8}$$

The last block stacks one scalar $\Delta\theta^{\mathrm{rob}}_{k,\mathrm{sys}}(\theta_k^{(l)})$ for every joint $k = 1, \ldots, 6$ and selected position $l = 0, \ldots, L_k$, ordered joint by joint. The vector $\boldsymbol{x}$ contains these finitely many values, not the distortion functions themselves. The prescribed positions $\theta_k^{(l)}$ are not entries of $\boldsymbol{x}$.

Table 6: Parameters of the calibrated model.

Parameters	Meaning	Count
$\boldsymbol{\alpha}_X$, $\boldsymbol{\epsilon}_X$	rotation and translation corrections to a fixed base reference, selected during initialization	6
$t_{k,1}$, $t_{k,2}$, $k = 2, \ldots, 6$	tilts $\boldsymbol{\phi}_k = t_{k,1}\boldsymbol{v}_{k,1} + t_{k,2}\boldsymbol{v}_{k,2}$, with unit vectors $\boldsymbol{v}_{k,1}, \boldsymbol{v}_{k,2} \perp \boldsymbol{d}_k$	10
$\boldsymbol{\alpha}_{\mathrm{we}}$, $\boldsymbol{\epsilon}_{\mathrm{we}}$	tool error ΔF_{we}	6
$\Delta\theta^{\mathrm{rob}}_{k,\mathrm{sys}}(\theta_k^{(l)})$	distortion of joint k at the selected encoder position $\theta_k^{(l)}$	$\sum_{k=1}^{6}(L_k+1)$

The base parameters $\boldsymbol{\alpha}_X$ and $\boldsymbol{\epsilon}_X$ describe a correction to a fixed reference transform. For a given trial vector $\boldsymbol{x}$, the base correction, axis tilts, tool error and distortion values $\Delta\theta^{\mathrm{rob}}_{k,\mathrm{sys}}(\theta_k^{(l)})$

stored in $\boldsymbol{x}$ are shared by all predictions. For example, the first two observations are compared with

$$F_{m,\mathrm{obs},1} \approx F_X(\boldsymbol{x})\hat{F}_{\mathrm{ee}}(\boldsymbol{\theta}_1;\boldsymbol{x}),$$
$$F_{m,\mathrm{obs},2} \approx F_X(\boldsymbol{x})\hat{F}_{\mathrm{ee}}(\boldsymbol{\theta}_2;\boldsymbol{x}).$$

The commanded joint values $\boldsymbol{\theta}_1$ and $\boldsymbol{\theta}_2$ differ, but the selected positions $\theta_k^{(l)}$ and the parameter vector $\boldsymbol{x}$ are the same in both predictions. The interpolation weights depend on the input joint values, so $\Delta\theta^{\mathrm{rob}}_{k,\mathrm{sys,est}}(\theta_{k,1})$ and $\Delta\theta^{\mathrm{rob}}_{k,\mathrm{sys,est}}(\theta_{k,2})$ can differ even though both are computed from the same fitted values $\Delta\theta^{\mathrm{rob}}_{k,\mathrm{sys,est}}(\theta_k^{(l)})$. We fit one $\boldsymbol{x}$ to all observed poses $F_{m,\mathrm{obs},j}$ in a single least-squares problem.

3.3 Calibration pipeline

Single-joint observations initialize the base, axes, tool and distortion values. One joint pose fit then refines these parameters; Section 3.4 analyses the approximations and remaining uncertainty.

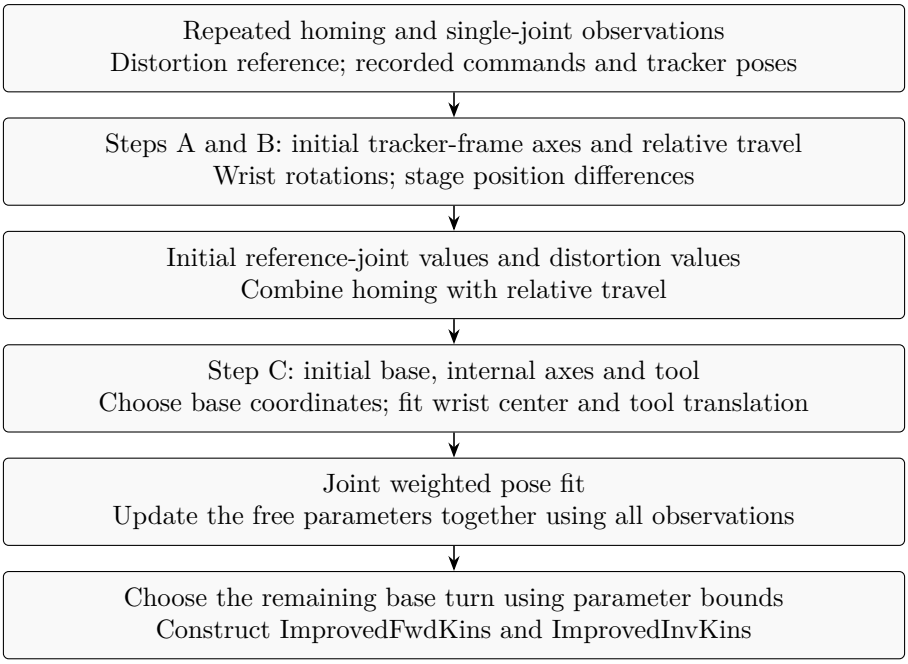


Figure 1: Initial estimates followed by joint parameter refinement.

3.3.1 Reference commands and averaged observations

Calibration reference. Record $F_{m,\mathrm{obs},0}$ at the zero encoder command:

$$\boldsymbol{\theta}_0 = \begin{bmatrix}0 & 0 & 0 & 0 & 0 & 0\end{bmatrix}^T, \qquad \theta^*_{i,0} = \Delta\theta^{\mathrm{rob}}_{i,\mathrm{sys}}(0) + \Delta\theta^{\mathrm{rob}}_{i,\mathrm{rnd},0}. \tag{9}$$

For joint k, sample $j \geq 1$ changes only that joint command:

$$\theta_{i,j} = \theta_{i,0} = 0, \qquad i \in \{1,\ldots,6\},\ i \neq k.$$

Repeated arrivals at the same command. Repeat each commanded configuration to reduce the random effects of both joint repeatability and tracker measurement. Between two arrivals, visit another configuration and include the full travel time:

$$\boldsymbol{\theta}_j \;\longrightarrow\; \boldsymbol{\theta}_{j'} \;\longrightarrow\; \boldsymbol{\theta}_j, \qquad \boldsymbol{\theta}_{j'} \neq \boldsymbol{\theta}_j.$$

Take one tracker observation at each arrival $q = 1, \ldots, n_j$. The command and systematic joint distortion are unchanged, but the joint random error is sampled again:

$$\theta_{i,j}^{*,(q)} = \theta_{i,j} + \Delta\theta_{i,\mathrm{sys}}^{\mathrm{rob}}(\theta_{i,j}) + \Delta\theta_{i,\mathrm{rnd},j}^{\mathrm{rob},(q)}.$$

The model treats the move errors as independent; fixed systematic errors are not removed by repetition.

Averaging observations at the same command. Average the n_j observed positions:

$$\boldsymbol{p}_{m,\mathrm{obs},j} = \frac{1}{n_j} \sum_{q=1}^{n_j} \boldsymbol{p}_{m,\mathrm{obs},j}^{(q)}.$$

The two SVD calculations have different inputs:

$$\underbrace{\{\boldsymbol{b}_{a,\mathrm{obs},j}^{(q)}\}_{a=1}^{N} \xrightarrow{\mathrm{SVD}} R_{m,\mathrm{obs},j}^{(q)}}_{\text{First SVD: marker registration}} \quad\longrightarrow\quad \underbrace{\{R_{m,\mathrm{obs},j}^{(q)}\}_{q=1}^{n_j} \xrightarrow{\mathrm{SVD}} R_{m,\mathrm{obs},j}}_{\text{Second SVD: rotation averaging}}.$$

The first estimates each arrival's pose from marker points. The second combines the resulting rotations into one rotation matrix. Use the orthonormal coordinate unit vectors

$$\boldsymbol{e}_x = \begin{bmatrix}1\\0\\0\end{bmatrix}, \qquad \boldsymbol{e}_y = \begin{bmatrix}0\\1\\0\end{bmatrix}, \qquad \boldsymbol{e}_z = \begin{bmatrix}0\\0\\1\end{bmatrix}.$$

For this average, fit the three observed coordinate-axis directions over rotation matrices R:

$$R_{m,\mathrm{obs},j} = \arg\min_{R} \sum_{q=1}^{n_j} \sum_{\boldsymbol{e}\in\{\boldsymbol{e}_x,\boldsymbol{e}_y,\boldsymbol{e}_z\}} \left\| R\boldsymbol{e} - R_{m,\mathrm{obs},j}^{(q)}\boldsymbol{e} \right\|^2.$$

Retain the number of arrivals as the weight of the averaged pose:

$$w_j = n_j.$$

Under the independent-move model, averaging reduces both joint and tracker random effects. The remaining uncertainty is analysed in Section 3.4.6.

3.3.2 Step A: initial wrist-axis directions and angular travel

Input and physical relation. For $k \in \{4, 5, 6\}$, use the reference rotation $R_{m,\mathrm{obs},0}$, the rotations $R_{m,\mathrm{obs},j}$ and encoder changes $\theta_{k,j} - \theta_{k,0}$. If all held physical wrist angles remain fixed, the true rotations satisfy

$$R_{m,j}^{*}(R_{m,0}^{*})^{T} = R\big(\boldsymbol{\ell}_k^{*}, \theta_{k,j}^{*} - \theta_{k,0}^{*}\big)\,. \tag{10}$$

Here $\boldsymbol{\ell}_k^{*}$ is the true unit axis in tracker coordinates at the reference configuration.

Rotation vectors from observations. Convert the observed relative rotation to axis-angle form. Its angle is computed from the trace:

$$\beta_j = \arccos\left(\frac{\operatorname{tr}(R_{m,\mathrm{obs},j} R_{m,\mathrm{obs},0}^T) - 1}{2}\right), \qquad 0 \le \beta_j \le \pi.$$

The range follows from arccos. The implementation clips its argument to $[-1, 1]$ to avoid roundoff errors. For $0 < \beta_j < \pi$, obtain the tracker-frame rotation vector, axis multiplied by angle, from

$$\operatorname{sk}(\boldsymbol{v}_j) = \frac{\beta_j}{2\sin\beta_j}\left(R_{m,\mathrm{obs},j} R_{m,\mathrm{obs},0}^T - R_{m,\mathrm{obs},0} R_{m,\mathrm{obs},j}^T\right), \qquad \|\boldsymbol{v}_j\|_2 = \beta_j.$$

For $\beta_j < 10^{-12}$, the implementation uses the small-angle form

$$\operatorname{sk}(\boldsymbol{v}_j) \approx \tfrac{1}{2}\left(R_{m,\mathrm{obs},j} R_{m,\mathrm{obs},0}^T - R_{m,\mathrm{obs},0} R_{m,\mathrm{obs},j}^T\right).$$

The implementation's reference-frame relative rotation gives the same rotation after a coordinate change:

$$R_{m,\mathrm{obs},j} R_{m,\mathrm{obs},0}^T = R_{m,\mathrm{obs},0}\big(R_{m,\mathrm{obs},0}^T R_{m,\mathrm{obs},j}\big) R_{m,\mathrm{obs},0}^T.$$

With the same axis-angle choice, multiplying its rotation vector by $R_{m,\mathrm{obs},0}$ gives $\boldsymbol{v}_j$. For direction initialization, approximate physical angular travel by encoder change:

$$\boldsymbol{v}_j \approx (\theta_{k,j} - \theta_{k,0})\boldsymbol{\ell}_k.$$

Weighted direction fit. Using the direction-fitting samples and visit-count weight w_j, search over unit directions $\boldsymbol{\ell}_k$. For each residual $\boldsymbol{e}_j$, weighting its square or scaling the residual are equivalent:

$$w_j \|\boldsymbol{e}_j\|_2^2 = \left\|\sqrt{w_j}\boldsymbol{e}_j\right\|_2^2, \qquad \|R_{m,\mathrm{obs},0}\boldsymbol{e}_j\|_2^2 = \|\boldsymbol{e}_j\|_2^2.$$

The coordinate change preserves the direction-fit objective. Thus

$$\boldsymbol{\ell}_{k,\mathrm{est}} = \arg\min_{\boldsymbol{\ell}_k} \sum_j w_j \|\boldsymbol{v}_j - (\theta_{k,j} - \theta_{k,0})\boldsymbol{\ell}_k\|_2^2. \tag{11}$$

Expanding the objective with $\|\boldsymbol{\ell}_k\|_2 = 1$ gives

$$\begin{aligned}
&\sum_j w_j \|\boldsymbol{v}_j - (\theta_{k,j} - \theta_{k,0})\boldsymbol{\ell}_k\|_2^2 \\
&\quad = \underbrace{\sum_j w_j\left[\|\boldsymbol{v}_j\|_2^2 + (\theta_{k,j} - \theta_{k,0})^2\right]}_{\text{independent of the candidate direction}} \\
&\qquad - 2\boldsymbol{\ell}_k^T \sum_j w_j(\theta_{k,j} - \theta_{k,0})\boldsymbol{v}_j.
\end{aligned}$$

Aligning the unit direction with the weighted sum gives

$$\boldsymbol{\ell}_{k,\mathrm{est}} = \frac{\sum_j w_j(\theta_{k,j} - \theta_{k,0})\boldsymbol{v}_j}{\left\|\sum_j w_j(\theta_{k,j} - \theta_{k,0})\boldsymbol{v}_j\right\|_2}.$$

The denominator must be nonzero. Encoder-change signs align positive and negative rotations to the same axis direction.

Signed travel and angle branches. For each sample of this joint, compute the signed angle in $[-\pi, \pi]$

$$\beta_{k,j} = \operatorname{sign}(\boldsymbol{v}_j^T \boldsymbol{\ell}_{k,\text{est}}) \left\| \boldsymbol{v}_j \right\|_2 .$$

Choose the 2π branch nearest the encoder change:

$$\Delta\theta_{k,j,\text{est}} = \beta_{k,j} + 2\pi \operatorname{round}\left(\frac{\theta_{k,j} - \theta_{k,0} - \beta_{k,j}}{2\pi} \right). \tag{12}$$

The outputs are the initial tracker-frame axis $\boldsymbol{\ell}_{k,\text{est}}$ and relative physical travel estimate $\Delta\theta_{k,j,\text{est}} \approx \theta^*_{k,j} - \theta^*_{k,0}$. Absolute physical angles are recovered after estimating the reference joint values.

3.3.3 Step B: initial stage directions and linear travel

Input and physical relation. For $k \in \{1, 2, 3\}$, use $\boldsymbol{p}_{m,\text{obs},0}$, $\boldsymbol{p}_{m,\text{obs},j}$ and the encoder changes. With all held physical joint values fixed, the wrist and tool positions relative to the translating carriage are unchanged. The true position difference is therefore

$$\boldsymbol{p}^*_{m,j} - \boldsymbol{p}^*_{m,0} = \underbrace{R^*_X \boldsymbol{d}^*_k}_{\boldsymbol{\ell}^*_k} (\theta^*_{k,j} - \theta^*_{k,0}). \tag{13}$$

Fit the observed position differences directly.

Weighted direction fit. For initialization, approximate physical travel by encoder change and fit a candidate unit direction:

$$\boldsymbol{\ell}_{k,\text{est}} = \arg\min_{\boldsymbol{\ell}_k} \sum_j w_j \left\| \boldsymbol{p}_{m,\text{obs},j} - \boldsymbol{p}_{m,\text{obs},0} - (\theta_{k,j} - \theta_{k,0}) \boldsymbol{\ell}_k \right\|^2. \tag{14}$$

The same expansion as in Step A leaves only a term linear in the candidate unit direction, giving

$$\boldsymbol{\ell}_{k,\text{est}} = \frac{\sum_j w_j (\theta_{k,j} - \theta_{k,0})(\boldsymbol{p}_{m,\text{obs},j} - \boldsymbol{p}_{m,\text{obs},0})}{\left\| \sum_j w_j (\theta_{k,j} - \theta_{k,0})(\boldsymbol{p}_{m,\text{obs},j} - \boldsymbol{p}_{m,\text{obs},0}) \right\|_2}.$$

Again, nonzero net direction information is needed. Project onto this fitted unit direction to estimate signed travel:

$$\Delta\theta_{k,j,\text{est}} = \boldsymbol{\ell}^T_{k,\text{est}} (\boldsymbol{p}_{m,\text{obs},j} - \boldsymbol{p}_{m,\text{obs},0}).$$

The projection estimates physical travel independently of the encoder-travel approximation used for direction initialization.

3.3.4 Initial reference-joint values and distortion values

Steps A and B estimate relative travel. Absolute physical joint values additionally need a reference, supplied by repeated homing.

Homing reference. At each homing h, the physical value of joint i is zero:

$$0 = \theta^{(h)}_{\text{encoders},i} + \Delta\theta^{\text{rob}}_{i,\text{sys}}(\theta^{(h)}_{\text{encoders},i}) + \Delta\theta^{\text{rob}}_{i,\text{rnd},h}, \qquad h = 1, \ldots, H.$$

Define the mean returned encoder value, then average the exact home equations:

$$\begin{aligned}
\bar{\theta}_{\text{encoders},i} &= \frac{1}{H}\sum_{h=1}^{H} \theta^{(h)}_{\text{encoders},i}, \\
-\bar{\theta}_{\text{encoders},i} &= \frac{1}{H}\sum_{h=1}^{H} \Delta\theta^{\text{rob}}_{i,\text{sys}}(\theta^{(h)}_{\text{encoders},i}) + \frac{1}{H}\sum_{h=1}^{H} \Delta\theta^{\text{rob}}_{i,\text{rnd},h}.
\end{aligned}$$

For independent zero-mean homing errors,

$$\mathbb{E}\left[\frac{1}{H}\sum_{h} \Delta\theta^{\text{rob}}_{i,\text{rnd},h}\right] = 0, \qquad \text{Std}\left(\frac{1}{H}\sum_{h} \Delta\theta^{\text{rob}}_{i,\text{rnd},h}\right) = \frac{\sigma_{i,\text{rnd}}}{\sqrt{H}}.$$

Finite averaging reduces this random term; it does not set it exactly to zero.

The average of the true distortion values is not generally the distortion at the average encoder value. The derivative bound gives

$$\begin{aligned}
&\left|\frac{1}{H}\sum_{h=1}^{H} \Delta\theta^{\text{rob}}_{i,\text{sys}}(\theta^{(h)}_{\text{encoders},i}) - \Delta\theta^{\text{rob}}_{i,\text{sys}}(\bar{\theta}_{\text{encoders},i})\right| \\
&\quad \le \frac{1}{H}\sum_{h=1}^{H} \left|\Delta\theta^{\text{rob}}_{i,\text{sys}}(\theta^{(h)}_{\text{encoders},i}) - \Delta\theta^{\text{rob}}_{i,\text{sys}}(\bar{\theta}_{\text{encoders},i})\right| \\
&\quad \le \frac{\omega^{\text{rob}}_{i,\text{sys}}}{H}\sum_{h=1}^{H} \left|\theta^{(h)}_{\text{encoders},i} - \bar{\theta}_{\text{encoders},i}\right|.
\end{aligned}$$

When this variation and the random mean are small, the true distortion satisfies

$$\Delta\theta^{\text{rob}}_{i,\text{sys}}(\bar{\theta}_{\text{encoders},i}) \approx -\bar{\theta}_{\text{encoders},i}.$$

This motivates defining the computable estimate

$$\Delta\theta^{\text{rob}}_{i,\text{sys,est}}(\bar{\theta}_{\text{encoders},i}) := -\bar{\theta}_{\text{encoders},i}. \tag{15}$$

Reference-joint initialization. Initially extend the estimated home distortion as a constant:

$$\Delta\theta^{\text{rob}}_{k,\text{sys,est}}(\theta_k) := \Delta\theta^{\text{rob}}_{k,\text{sys,est}}(\bar{\theta}_{\text{encoders},k}) = -\bar{\theta}_{\text{encoders},k}.$$

Substitute this initial model into the reference-joint calculation:

$$\theta_{k,0,\text{est}} = \theta_{k,0} + \Delta\theta^{\text{rob}}_{k,\text{sys,est}}(\theta_{k,0}) = \theta_{k,0} - \bar{\theta}_{\text{encoders},k}.$$

The true distortion is not assumed constant. For each observed joint position, compute

$$\begin{aligned}
\theta_{k,j,\text{est}} &= \theta_{k,0,\text{est}} + \Delta\theta_{k,j,\text{est}}, \\
y_{k,j} &= \theta_{k,j,\text{est}} - \theta_{k,j}.
\end{aligned} \tag{16}$$

These $y_{k,j}$ initialize the distortion values at the selected encoder positions by linear interpolation. Include the reference command and its initial distortion in this interpolation.

Replace the selected encoder position nearest $\bar{\theta}_{\text{encoders},k}$ by that mean home reading, and fix the distortion there:

$$\Delta\theta^{\text{rob}}_{k,\text{sys,est}}(\bar{\theta}_{\text{encoders},k}) = -\bar{\theta}_{\text{encoders},k}.$$

The other selected distortion values remain free. Default spacings are 100 mm and $\pi/4$ rad; the model interpolates linearly inside the selected range and extrapolates linearly outside it.

3.3.5 Step C: initial base orientation and internal axis directions

Input. Use the six tracker-frame axis estimates and the reference-joint estimates. Here $\boldsymbol{e}_x, \boldsymbol{e}_y, \boldsymbol{e}_z$ denote the orthonormal robot-base coordinate directions; the physical joint axes $\boldsymbol{d}^*_k$ may be tilted relative to them. For stages the physical relation is $\boldsymbol{\ell}^*_k = R^*_X \boldsymbol{d}^*_k$. The known $\boldsymbol{d}^*_1 = \boldsymbol{e}_x$ fixes the base x direction. Choosing a base y direction along the perpendicular component of $\boldsymbol{\ell}_{2,\text{est}}$ gives

$$\begin{aligned}
R_{X,\text{est}}\boldsymbol{e}_x &= \boldsymbol{\ell}_{1,\text{est}}, \\
R_{X,\text{est}}\boldsymbol{e}_y &= \frac{(I - \boldsymbol{\ell}_{1,\text{est}}\boldsymbol{\ell}^T_{1,\text{est}})\boldsymbol{\ell}_{2,\text{est}}}{\left\|(I - \boldsymbol{\ell}_{1,\text{est}}\boldsymbol{\ell}^T_{1,\text{est}})\boldsymbol{\ell}_{2,\text{est}}\right\|_2}, \\
R_{X,\text{est}}\boldsymbol{e}_z &= (R_{X,\text{est}}\boldsymbol{e}_x) \times (R_{X,\text{est}}\boldsymbol{e}_y).
\end{aligned}$$

The construction imposes a coordinate choice:

$$\boldsymbol{d}^T_{2,\text{est}}\boldsymbol{e}_z = 0, \qquad \boldsymbol{\phi}^T_2\boldsymbol{e}_x = 0 \quad \text{during the fit.}$$

It does not imply $(\boldsymbol{d}^*_2)^T\boldsymbol{e}_z = 0$. The unmeasured base turn is selected from parameter bounds after fitting.

Why wrist reference angles must be removed. The internal wrist axes are defined before the preceding wrist rotations. For example, at the reference,

$$\boldsymbol{\ell}^*_4 = R^*_X\boldsymbol{d}^*_4, \qquad \boldsymbol{\ell}^*_5 = R^*_X R(\boldsymbol{d}^*_4, \theta^*_{4,0})\boldsymbol{d}^*_5,$$

$$\boldsymbol{\ell}^*_6 = R^*_X R(\boldsymbol{d}^*_4, \theta^*_{4,0}) R(\boldsymbol{d}^*_5, \theta^*_{5,0})\boldsymbol{d}^*_6.$$

Thus remove preceding reference rotations in the order $4, 5, 6$:

$$\begin{aligned}
\boldsymbol{d}_{k,\text{est}} &= R^T_{X,\text{est}}\boldsymbol{\ell}_{k,\text{est}}, && k = 1, 2, 3, 4, \\
\boldsymbol{d}_{5,\text{est}} &= R(\boldsymbol{d}_{4,\text{est}}, \theta_{4,0,\text{est}})^T R^T_{X,\text{est}}\boldsymbol{\ell}_{5,\text{est}}, \\
\boldsymbol{d}_{6,\text{est}} &= R(\boldsymbol{d}_{5,\text{est}}, \theta_{5,0,\text{est}})^T R(\boldsymbol{d}_{4,\text{est}}, \theta_{4,0,\text{est}})^T R^T_{X,\text{est}}\boldsymbol{\ell}_{6,\text{est}}.
\end{aligned}$$

The estimated directions must be normalized. The code converts them to the shortest tilt vectors mapping nominal axes to estimated axes. For $k = 2, \ldots, 6$,

$$\boldsymbol{\phi}_{k,\text{est}} = \frac{\boldsymbol{d}_k \times \boldsymbol{d}_{k,\text{est}}}{\|\boldsymbol{d}_k \times \boldsymbol{d}_{k,\text{est}}\|_2} \operatorname{atan2}\left(\|\boldsymbol{d}_k \times \boldsymbol{d}_{k,\text{est}}\|_2, \boldsymbol{d}^T_k\boldsymbol{d}_{k,\text{est}}\right).$$

Use $\boldsymbol{\phi}_{k,\text{est}} = \boldsymbol{0}$ when $\|\boldsymbol{d}_k \times \boldsymbol{d}_{k,\text{est}}\|_2 < 10^{-15}$. This implementation threshold assumes nearby, not antiparallel, axes.

3.3.6 Step C: initial tool transform, wrist center and base position

Tool rotation from the reference observation. Construct the reference wrist rotation from the axes just obtained:

$$R_{w,0,\mathrm{est}} = R(\boldsymbol{d}_{4,\mathrm{est}}, \theta_{4,0,\mathrm{est}})R(\boldsymbol{d}_{5,\mathrm{est}}, \theta_{5,0,\mathrm{est}})R(\boldsymbol{d}_{6,\mathrm{est}}, \theta_{6,0,\mathrm{est}}).$$

Since F_{cw} and nominal F_{we} have identity rotations, the reference relation and its estimated counterpart are

$$R^*_{m,0} = R^*_X R^*_{w,0} R^*_{\mathrm{we}}, \qquad R_{\mathrm{we,est}} = R^T_{w,0,\mathrm{est}} R^T_{X,\mathrm{est}} R_{m,\mathrm{obs},0}.$$

Joint linear fit for wrist center and tool translation. Use all wrist observations and include the reference once. Let $\boldsymbol{o}_w$ be the wrist center in tracker coordinates when the physical stages retain their reference values. Let $\boldsymbol{s}$ be the vector from the wrist center to the end-effector origin, expressed in the end-effector frame. It is constant for a rigid tool. The actual geometry gives

$$\boldsymbol{p}^*_{m,j} = \boldsymbol{o}_w + R^*_{m,j}\boldsymbol{s}, \qquad \boldsymbol{s} = (R^*_{\mathrm{we}})^T \boldsymbol{p}^*_{\mathrm{we}}.$$

Substitute observed pose blocks and stack the weighted residuals:

$$\begin{bmatrix} \boldsymbol{s}_{\mathrm{est}} \\ \boldsymbol{o}_{w,\mathrm{est}} \end{bmatrix} = \arg\min_{\boldsymbol{s},\boldsymbol{o}} \left\| \begin{bmatrix} \vdots \\ \sqrt{w_j}[R_{m,\mathrm{obs},j} \quad I] \\ \vdots \end{bmatrix} \begin{bmatrix} \boldsymbol{s} \\ \boldsymbol{o} \end{bmatrix} - \begin{bmatrix} \vdots \\ \sqrt{w_j}\boldsymbol{p}_{m,\mathrm{obs},j} \\ \vdots \end{bmatrix} \right\|^2 . \tag{17}$$

Stack joints $4, 5, 6$ and the reference once. Nonzero rotations about at least two nonparallel axes are needed for a unique center.

Convert the fitted tool vector to the wrist frame and subtract nominal translation:

$$\boldsymbol{p}_{\mathrm{we,est}} = R_{\mathrm{we,est}}\boldsymbol{s}_{\mathrm{est}}, \qquad \boldsymbol{\epsilon}_{\mathrm{we,est}} = \boldsymbol{p}_{\mathrm{we,est}} - \boldsymbol{p}_{\mathrm{we}}, \qquad R(\boldsymbol{\alpha}_{\mathrm{we,est}}) = R_{\mathrm{we,est}}.$$

At the reference the carriage translation is generally nonzero. Therefore

$$\begin{aligned} \boldsymbol{p}_{X,\mathrm{est}} &= \boldsymbol{o}_{w,\mathrm{est}} - R_{X,\mathrm{est}} \left(\sum_{i=1}^{3} \theta_{i,0,\mathrm{est}} \boldsymbol{d}_{i,\mathrm{est}} + \boldsymbol{p}_{\mathrm{cw}} \right), \\ F_{X,\mathrm{est}} &= \mathrm{Frame}(R_{X,\mathrm{est}}, \boldsymbol{p}_{X,\mathrm{est}}). \end{aligned}$$

3.3.7 Packing the initial free parameters

With $\boldsymbol{\phi}_2^T \boldsymbol{e}_x = 0$ and one home distortion value fixed per joint, the free-parameter counts are

	base	axis tilts	tool	distortion values
count	6	9	6	$\sum_{k=1}^{6} L_k$

.

The assignment PDF does not prescribe the reference transform used for the base correction. One choice is the known nominal base $F_X = F^{-1}_{\mathrm{tracker}} F_{\mathrm{rob}}$. The course-code template instead suggests estimating the base from the observations with nominal kinematics. We follow that suggestion here; no accuracy advantage of either choice is assumed.

The nominal-model relation gives

$$F_{m,\mathrm{obs},j} \approx F_X F_{\mathrm{ee}}(\boldsymbol{\theta}_j) \quad \Longrightarrow \quad F_{m,\mathrm{obs},j} F_{\mathrm{ee}}(\boldsymbol{\theta}_j)^{-1} \approx F_X.$$

Each observation supplies a base estimate. Combine these estimates using the position mean and rotation least squares described above:

$$F_{X,\text{init}} = \text{AverageFrames}_j \left(F_{m,\text{obs},j} F_{\text{ee}}(\boldsymbol{\theta}_j)^{-1} \right) = \text{Frame}(R_{X,\text{init}}, \boldsymbol{p}_{X,\text{init}}).$$

The resulting $F_{X,\text{init}}$ is held fixed as the correction reference; it can still contain effects of nominal-kinematics errors. The adjustable base pose is

$$F_X(\boldsymbol{x}) = F_{X,\text{init}} \, \text{Frame}(R(\boldsymbol{\alpha}_X), \boldsymbol{\epsilon}_X) \, .$$

To reproduce the base estimate $F_{X,\text{est}}$ obtained above, initialize the correction parameters by

$$\begin{aligned} \text{Frame}(R(\boldsymbol{\alpha}_X), \boldsymbol{\epsilon}_X) &= F_{X,\text{init}}^{-1} F_{X,\text{est}}, \\ R(\boldsymbol{\alpha}_X) &= R_{X,\text{init}}^T R_{X,\text{est}}, \\ \boldsymbol{\epsilon}_X &= R_{X,\text{init}}^T (\boldsymbol{p}_{X,\text{est}} - \boldsymbol{p}_{X,\text{init}}). \end{aligned}$$

Initialization detail	Setting
Repeated homings	$H = 5$ by default
Wrist-axis sample filter	Computed rotation-vector norm $< 2.8\,\text{rad}$
Angular-travel calculation	All wrist samples
Single-joint initialization	Recorded reference and at least two stops per joint
Missing initialization data	Fall back to the current parameter vector
Optional initialization mode	Keep current parameters without the single-joint calculation
Stored distortion parameters	Deviations from the measured home distortion value
Repeat order (report / current code)	Separated visits / consecutive commands

Table 7: Initialization settings.

The wrist-axis filter avoids rotation-vector axis ambiguity near π. Its 2.8 rad threshold is an implementation choice, not a derived accuracy condition; it does not exclude observations from the angular-travel calculation.

3.3.8 Joint refinement using all observed poses

Prediction and residual. For each recorded $\boldsymbol{\theta}_j$, evaluate the piecewise-linear distortion and the complete kinematics to form

$$\hat{F}_{m,j}(\boldsymbol{x}) = F_X(\boldsymbol{x}) \hat{F}_{\text{ee}}(\boldsymbol{\theta}_j; \boldsymbol{x}) = \text{Frame}(\hat{R}_{m,j}(\boldsymbol{x}), \hat{\boldsymbol{p}}_{m,j}(\boldsymbol{x})).$$

Use modeled physical joint values

$$\theta_{k,\text{est}} = \theta_k + \Delta\theta_{k,\text{sys,est}}^{\text{rob}}(\theta_k).$$

Compare the prediction with the observed pose:

$$\begin{aligned} \hat{F}_{m,j}(\boldsymbol{x})^{-1} F_{m,\text{obs},j} &= \text{Frame}(\exp[\text{sk}(\boldsymbol{\alpha}_j)], \boldsymbol{\epsilon}_j) \, , \\ \exp[\text{sk}(\boldsymbol{\alpha}_j)] &= \hat{R}_{m,j}^T R_{m,\text{obs},j}, \\ \boldsymbol{\epsilon}_j &= \hat{R}_{m,j}^T (\boldsymbol{p}_{m,\text{obs},j} - \hat{\boldsymbol{p}}_{m,j}). \end{aligned} \tag{18}$$

Convert $\hat{R}_{m,j}^T R_{m,\text{obs},j}$ to the rotation vector $\boldsymbol{\alpha}_j$, choosing $0 \le \|\boldsymbol{\alpha}_j\|_2 \le \pi$. Minimize the weighted rotation and translation errors over all observations:

$$\boldsymbol{x}_{\text{est}} = \arg\min_{\boldsymbol{x}} \sum_j w_j \left[\frac{\|\boldsymbol{\alpha}_j(\boldsymbol{x})\|_2^2}{\sigma_R^2} + \frac{\|\boldsymbol{\epsilon}_j(\boldsymbol{x})\|_2^2}{\sigma_p^2} \right]. \tag{19}$$

Marker layout and residual scales. The scales σ_R, σ_p normalize radians and millimetres and set their relative weights. To choose them, propagate marker random noise through rigid registration, using the same linearization as in Assignment 1. The A2 calibration body has exact local marker coordinates $\boldsymbol{a}_k^*$, so no marker-coordinate calibration error is included here.

Choose four markers at the vertices of a regular tetrahedron centred at the end-effector origin:

$$\begin{bmatrix}\boldsymbol{a}_1^* & \boldsymbol{a}_2^* & \boldsymbol{a}_3^* & \boldsymbol{a}_4^*\end{bmatrix} = \frac{r}{\sqrt{3}}\begin{bmatrix}1 & 1 & -1 & -1\\ 1 & -1 & 1 & -1\\ 1 & -1 & -1 & 1\end{bmatrix}.$$

Then

$$\sum_{k=1}^{4}\boldsymbol{a}_k^* = \mathbf{0}, \qquad \|\boldsymbol{a}_k^*\|_2 = r, \qquad \sum_{k=1}^{4}\boldsymbol{a}_k^*(\boldsymbol{a}_k^*)^T = \frac{4r^2}{3}I.$$

For a small pose error, the change in a marker position is linear in the rotation and translation errors, with coefficient matrix $R[-\operatorname{sk}(\boldsymbol{a}_k^*) \quad I]$. Since $R^TR = I$, the normal matrix for all four markers is

$$\sum_{k=1}^{4}\begin{bmatrix}-\operatorname{sk}(\boldsymbol{a}_k^*) & I\end{bmatrix}^T\begin{bmatrix}-\operatorname{sk}(\boldsymbol{a}_k^*) & I\end{bmatrix} = \begin{bmatrix}G & 0\\ 0 & 4I\end{bmatrix},$$

$$G = \sum_{k=1}^{4}\left(\|\boldsymbol{a}_k^*\|_2^2\, I - \boldsymbol{a}_k^*(\boldsymbol{a}_k^*)^T\right) = \frac{8r^2}{3}I.$$

The zero off-diagonal blocks follow from the zero centroid. With independent marker noise of covariance $\sigma_b^2 I$, the linear least-squares pose covariance is therefore

$$\sigma_b^2\begin{bmatrix}G & 0\\ 0 & 4I\end{bmatrix}^{-1} = \begin{bmatrix}\dfrac{3\sigma_b^2}{8r^2}I & 0\\ 0 & \dfrac{\sigma_b^2}{4}I\end{bmatrix}.$$

Thus the tetrahedron gives the same rotational precision about all three axes. For four centred markers within radius r, $\operatorname{tr}(G) \le 8r^2$, hence $\lambda_{\min}(G) \le 8r^2/3$; the tetrahedron attains this bound. This justifies its geometry for the chosen marker count, without asserting that four markers are optimal among all counts.

The handout permits $\|\boldsymbol{a}_k^*\|_2 \le 75\,\text{mm}$. Increasing r reduces rotation variance for this layout, so choose $r = 74.9\,\text{mm}$, just inside the limit. With the handout's $\sigma_b = 0.1\,\text{mm}$, the per-component standard deviations for one registration are

$$\sigma_p = \frac{\sigma_b}{\sqrt{4}} = 0.05\,\text{mm},$$

$$\sigma_R = \frac{\sigma_b}{r}\sqrt{\frac{3}{8}} \approx 8.2\times10^{-4}\,\text{rad}.$$

A rotation residual of magnitude σ_R and a translation residual of magnitude σ_p therefore contribute equally before multiplication by w_j. These are weighting scales, not calibration-accuracy targets. They describe marker random noise for this layout, not the complete residual covariance including joint errors and tracker systematic errors.

Joint iterations. For the numerical iteration, collect the same normalized residuals into a vector:

$$\boldsymbol{r}(\boldsymbol{x}) = \begin{bmatrix} \vdots \\ \sqrt{w_j}\boldsymbol{\alpha}_j(\boldsymbol{x})/\sigma_R \\ \sqrt{w_j}\boldsymbol{\epsilon}_j(\boldsymbol{x})/\sigma_p \\ \vdots \end{bmatrix}.$$

Its squared norm equals the objective above. With the current parameter vector, linearize:

$$\boldsymbol{r}(\boldsymbol{x}+\boldsymbol{\delta}) \approx \boldsymbol{r}(\boldsymbol{x}) + J\boldsymbol{\delta}, \qquad J = \frac{\partial \boldsymbol{r}}{\partial \boldsymbol{x}}.$$

Minimize the linearized residual with a trust-region step, then update all free parameters together:

$$\boldsymbol{\delta}: \ \min \|\boldsymbol{r} + J\boldsymbol{\delta}\|_2^2 \quad \text{with step-size control}, \qquad \boldsymbol{x} \leftarrow \boldsymbol{x} + \boldsymbol{\delta}.$$

Recompute the complete predictions after each update.

Joint-fit setting	Implementation value
Solver	Trust-region reflective least squares
Parameter scaling	Jacobian-column scaling
Cost, step and gradient tolerances	10^{-12}
Maximum function evaluations	200
Bounds passed to this solver	None
Allowed observations	Single-joint and simultaneous multi-joint motion

Table 8: Numerical settings for the joint pose fit.

3.3.9 Selecting the remaining base turn

The fit has used a coordinate choice for the base turn about $\boldsymbol{e}_x$. To see why observations cannot choose this turn directly, define $Q(c) = \mathrm{Frame}(R(\boldsymbol{e}_x, c), \mathbf{0})$. Then

$$F_X \hat{F}_{\mathrm{ee}}(\boldsymbol{\theta}) = \big(F_X Q(c)\big)\big(Q(-c)\hat{F}_{\mathrm{ee}}(\boldsymbol{\theta})\big). \tag{20}$$

Select c using the axis-tilt, tool-rotation and base-placement bounds; the pose residual is unchanged.

For each candidate c, rotate the fitted axes and tool by $Q(-c)$ and the base by $Q(c)$. The base-placement error relative to nominal workspace placement is

$$\Delta F_{\mathrm{rob}}(c) = F_{\mathrm{tracker}} F_{X,\mathrm{est}} Q(c) F_{\mathrm{rob}}^{-1}.$$

Let $\boldsymbol{\alpha}_{\mathrm{we}}(c)$ and $\boldsymbol{\alpha}_{\mathrm{rob}}(c)$ be the rotation vectors of $Q(-c)\Delta F_{\mathrm{we,est}}$ and $\Delta F_{\mathrm{rob}}(c)$. The transformed tilt angles satisfy

$$\|\boldsymbol{\phi}_k(c)\|_2 = \arccos\big(\boldsymbol{d}_k^T R(\boldsymbol{e}_x, -c)\boldsymbol{d}_{k,\mathrm{est}}\big).$$

Measure the largest normalized bound violation by

$$\rho(c) = \max\left\{ \max_{k\in\{2,3,5,6\}} \frac{\|\boldsymbol{\phi}_k(c)\|_2}{\phi_{\max}}, \ \frac{\|\boldsymbol{\alpha}_{\mathrm{we}}(c)\|_2}{0.01}, \ \frac{\|\boldsymbol{\alpha}_{\mathrm{rob}}(c)\|_2}{\alpha_{\mathrm{rob}}^{\max}} \right\}.$$

Include the base-error term only when the exact tracker placement is supplied. Axis $1, 4$ tilt magnitudes are invariant under this turn.

The implementation uses the grid $[-0.1, 0.1]$ rad with spacing 2×10^{-5} rad. Let $\mathcal{C}$ contain its accepted points:

$$\mathcal{C} = \{c \text{ on the grid} : \rho(c) \le 1\},$$
$$c_{\text{est}} = \begin{cases} \frac{1}{2}(\min \mathcal{C} + \max \mathcal{C}), & \mathcal{C} \neq \varnothing, \\ \underset{c \text{ on the grid}}{\arg\min}\ \rho(c), & \mathcal{C} = \varnothing. \end{cases}$$

The range, spacing and fallback are implementation choices. A midpoint minimizes the largest distance to the accepted endpoints; it does not measure the true turn.

3.3.10 Using the fitted model for forward and inverse kinematics

Forward prediction. Evaluate the full chain with fitted tilts, fitted tool error and interpolated distortion values, and apply the selected coordinate turn:

$$\text{ImprovedFwdKins}(\boldsymbol{\theta}) = Q(-c)\hat{F}_{\text{ee}}(\boldsymbol{\theta}).$$

Here $\hat{F}_{\text{ee}}$ uses the fitting coordinates. Return the base $F_{X,\text{est}}Q(c)$ so that

$$\big(F_{X,\text{est}}Q(c)\big)\,\text{ImprovedFwdKins}(\boldsymbol{\theta}) = F_{X,\text{est}}\hat{F}_{\text{ee}}(\boldsymbol{\theta}).$$

Inverse solution. For target F in the returned base coordinates, form the pose residual

$$\text{ImprovedFwdKins}(\boldsymbol{\theta})^{-1}F = \text{Frame}(\exp[\text{sk}(\boldsymbol{\alpha})], \boldsymbol{\epsilon}), \qquad \boldsymbol{g}(\boldsymbol{\theta}) = \begin{bmatrix} \boldsymbol{\alpha} \\ \boldsymbol{\epsilon} \end{bmatrix}.$$

Start from each nominal wrist branch, with nominal stage coordinates, and refine by a damped least-squares step:

$$\begin{aligned} \boldsymbol{\delta} &= -\left[J^T J + \lambda\,\text{diag}\big(\text{diag}(J^T J) + 10^{-12}\mathbf{1}\big)\right]^{-1} J^T \boldsymbol{g}, \\ \boldsymbol{\theta} &\leftarrow \boldsymbol{\theta} + \text{clip}(\boldsymbol{\delta}, -\boldsymbol{\delta}_{\max}, \boldsymbol{\delta}_{\max}), \qquad J = \frac{\partial \boldsymbol{g}}{\partial \boldsymbol{\theta}}. \end{aligned} \tag{21}$$

Inverse-solver setting	Implementation value
Jacobian forward-difference steps	10^{-3} mm; 10^{-6} rad
Componentwise step limits	50 mm; 0.3 rad
Initial damping	$\lambda = 10^{-4}$
Rejected trial	$\lambda \leftarrow 10\lambda$
Accepted trial	$\lambda \leftarrow \max(\lambda/10, 10^{-12})$
Damping-loop exit	Residual decreases or $\lambda > 10^6$
Maximum iterations per branch	50
Residual-norm tolerance	10^{-10}
Angle wrapping	Only for an out-of-limit candidate; verify the resulting pose

Table 9: Numerical controls for inverse kinematics.

Return the converged in-limit branch nearest the supplied seed, using the current command as the seed when none is supplied:

$$\boldsymbol{\theta}_{\text{return}} = \underset{\substack{\text{refined in-limit} \\ \text{nominal wrist branches}}}{\arg\min} \|\boldsymbol{\theta} - \boldsymbol{\theta}_{\text{seed}}\|_2 .$$

Check $\|\boldsymbol{g}(\boldsymbol{\theta})\|_2 \le \text{tol}$ after any angle wrapping and reject candidates that fail. Preserve already in-limit encoder angles because their fitted distortions need not be 2π-periodic. Raise an error if no candidate satisfies both limits and the residual tolerance.

3.4 Error analysis and parameter uncertainty

The following analysis distinguishes exact cancellations, random-noise reduction and errors retained in the fitted parameters.

3.4.1 What each operation removes or retains

Operation	Effect used in the pipeline	Remaining error or limitation
Relative wrist rotation	Cancels the common tool rotation and subsequent held rotations	Held wrist changes and tracker errors remain; principal angles have branches
Stage position difference	Cancels the fixed wrist/tool position when held physical joints are fixed	Held-joint changes and tracker position errors remain
Repeated homing	Averages independent homing errors to estimate an absolute distortion value	Finite-mean error and distortion variation are retained in the fixed home estimate
Repeated arrivals at one command	Reduces independent joint and tracker random components	Fixed systematic errors and shared-reference uncertainty remain
Initial encoder-travel approximation	Produces a simple direction fit	Does not estimate physical travel or systematic distortion exactly
Joint pose fit	Adjusts base, axes, tool and distortion values to reduce observed residuals	Noise, unmodelled tracker distortion, interpolation error and weakly determined parameters remain
Base-turn convention and bounds	Selects one representation of an observationally indistinguishable model	Does not recover a unique true base turn from tracker data

Table 10: Exact cancellations, statistical reductions and approximations.

3.4.2 Homing and nonzero reference-joint uncertainty

The defined home estimate has the exact error

$$\begin{aligned}
&\Delta\theta^{\mathrm{rob}}_{k,\mathrm{sys,est}}(\bar{\theta}_{\mathrm{encoders},k}) - \Delta\theta^{\mathrm{rob}}_{k,\mathrm{sys}}(\bar{\theta}_{\mathrm{encoders},k}) \\
&\quad = \frac{1}{H}\sum_{h=1}^{H}\left[\Delta\theta^{\mathrm{rob}}_{k,\mathrm{sys}}(\theta^{(h)}_{\mathrm{encoders},k}) - \Delta\theta^{\mathrm{rob}}_{k,\mathrm{sys}}(\bar{\theta}_{\mathrm{encoders},k})\right] \\
&\qquad + \frac{1}{H}\sum_{h=1}^{H}\Delta\theta^{\mathrm{rob}}_{k,\mathrm{rnd},h}.
\end{aligned}$$

Using the handout's derivative bound, the error in fixing the distortion at the mean home encoder position satisfies

$$\begin{aligned}
&\left|\Delta\theta^{\mathrm{rob}}_{k,\mathrm{sys}}(\bar{\theta}_{\mathrm{encoders},k}) + \bar{\theta}_{\mathrm{encoders},k}\right| \\
&\quad \le \left|\frac{1}{H}\sum_{h=1}^{H}\Delta\theta^{\mathrm{rob}}_{k,\mathrm{rnd},h}\right| + \frac{\omega^{\mathrm{rob}}_{k,\mathrm{sys}}}{H}\sum_{h=1}^{H}\left|\theta^{(h)}_{\mathrm{encoders},k} - \bar{\theta}_{\mathrm{encoders},k}\right|.
\end{aligned}$$

For independent homings,

$$\mathrm{Std}\left(\frac{1}{H}\sum_{h}\Delta\theta^{\mathrm{rob}}_{k,\mathrm{rnd},h}\right) = \frac{\sigma_{k,\mathrm{rnd}}}{\sqrt{H}}.$$

The derivative-bound term accounts for distortion variation between home readings. Both contributions remain when the estimated home value is fixed in the fit.

For any reference command, the physical value and the modeled estimate differ by

$$\begin{aligned}\theta_{k,0,\mathrm{est}} - \theta^*_{k,0} = \Delta\theta^{\mathrm{rob}}_{k,\mathrm{sys,est}}(\theta_{k,0}) - \Delta\theta^{\mathrm{rob}}_{k,\mathrm{sys}}(\theta_{k,0}) \\ - \Delta\theta^{\mathrm{rob}}_{k,\mathrm{rnd},0}.\end{aligned}$$

For the constant home-based initialization,

$$\begin{aligned}&\left|\Delta\theta^{\mathrm{rob}}_{k,\mathrm{sys,est}}(\theta_{k,0}) - \Delta\theta^{\mathrm{rob}}_{k,\mathrm{sys}}(\theta_{k,0})\right| \\ &\quad\le \left|\Delta\theta^{\mathrm{rob}}_{k,\mathrm{sys,est}}(\bar\theta_{\mathrm{encoders},k}) - \Delta\theta^{\mathrm{rob}}_{k,\mathrm{sys}}(\bar\theta_{\mathrm{encoders},k})\right| \\ &\qquad + \omega^{\mathrm{rob}}_{k,\mathrm{sys}}\left|\theta_{k,0} - \bar\theta_{\mathrm{encoders},k}\right|.\end{aligned}$$

Joint refinement adjusts free distortion values; the home value remains fixed.

3.4.3 Held-joint changes and shared reference errors

For a held joint $i \neq k$, compare one reference move with a later move at the same encoder command. Subtracting the two physical-joint equations cancels the systematic distortion exactly:

$$\delta\theta_{i,j} \equiv \theta^*_{i,j} - \theta^*_{i,0} = \Delta\theta^{\mathrm{rob}}_{i,\mathrm{rnd},j} - \Delta\theta^{\mathrm{rob}}_{i,\mathrm{rnd},0}. \tag{22}$$

For independent move errors,

$$\begin{aligned}\mathbb{E}[\delta\theta_{i,j}] = 0, \qquad \mathrm{Var}(\delta\theta_{i,j}) = 2\sigma^2_{i,\mathrm{rnd}}, \\ \mathrm{Cov}(\delta\theta_{i,j}, \delta\theta_{i,j'}) = \sigma^2_{i,\mathrm{rnd}}, \qquad j \neq j'.\end{aligned} \tag{23}$$

The reference error is shared. Conditional on it,

$$\mathbb{E}[\delta\theta_{i,j} \mid \Delta\theta^{\mathrm{rob}}_{i,\mathrm{rnd},0}] = -\Delta\theta^{\mathrm{rob}}_{i,\mathrm{rnd},0}.$$

For a reference averaged over independent moves, the shared covariance becomes the variance of that reference mean:

$$\mathrm{Cov}(\delta\theta_{i,j}, \delta\theta_{i,j'}) = \mathrm{Var}(\text{reference joint-error mean}), \qquad j \neq j'.$$

To quantify the local pose effect, let $\boldsymbol{r}^*_{m,j}$ be the tracker-frame vector from wrist center to end-effector origin, and let

$$\boldsymbol{u}^*_{i,j} = R^*_X \left(\prod_{h=4}^{i-1} R(\boldsymbol{d}^*_h, \theta^*_{h,j})\right) \boldsymbol{d}^*_i$$

be the actual tracker-frame wrist-axis direction at sample j. The product contains only rotations before axis i; for $i = 4$ it is I. For the right-multiplied pose variation expressed in end-effector coordinates,

$$\begin{bmatrix}\boldsymbol{\alpha}_{\mathrm{hold},j} \\ \boldsymbol{\epsilon}_{\mathrm{hold},j}\end{bmatrix} \approx \sum_{i\neq k} A_{i,j}\delta\theta_{i,j},$$

$$A_{i,j} = \begin{cases}\begin{bmatrix}\mathbf{0} \\ (R^*_{m,j})^T R^*_X \boldsymbol{d}^*_i\end{bmatrix}, & i \le 3, \\ \begin{bmatrix}(R^*_{m,j})^T \boldsymbol{u}^*_{i,j} \\ (R^*_{m,j})^T(\boldsymbol{u}^*_{i,j} \times \boldsymbol{r}^*_{m,j})\end{bmatrix}, & i \ge 4.\end{cases} \tag{24}$$

Keep the active physical value fixed. Evaluating $A_{i,j}$ at the actual configuration changes only second-order terms; the first-order covariance is

$$C_{\text{hold},j} \approx 2 \sum_{i \neq k} \sigma_{i,\text{rnd}}^2 A_{i,j} A_{i,j}^T. \tag{25}$$

The handout's tool error bound gives a wrist-to-tool distance at most 100.5 mm. With m_t held stages and m_r held wrist joints,

$$\begin{aligned} \sqrt{\mathbb{E}\,\|\boldsymbol{\alpha}_{\text{hold},j}\|_2^2} &\lesssim \sqrt{2m_r}\, 10^{-4} \text{ rad}, \\ \sqrt{\mathbb{E}\,\|\boldsymbol{\epsilon}_{\text{hold},j}\|_2^2} &\lesssim \sqrt{2m_t(0.005)^2 + 2m_r(100.5 \cdot 10^{-4})^2} \text{ mm}. \end{aligned} \tag{26}$$

Taking $m_t, m_r \leq 3$ gives first-order RMS scales 2.45×10^{-4} rad and 0.0275 mm. These are single-arrival pose-change scales, not deterministic bounds or accuracy bounds for the estimated axes.

3.4.4 Axis initialization, angle choices and travel errors

Step A. Write $\Delta\theta_{k,j}^* = \theta_{k,j}^* - \theta_{k,0}^*$. On a consistent angle branch, the rotation-vector residual at the true tracker-frame axis separates exactly as

$$\begin{aligned} \boldsymbol{v}_j - (\theta_{k,j} - \theta_{k,0})\boldsymbol{\ell}_k^* = & \underbrace{\boldsymbol{v}_j - \Delta\theta_{k,j}^* \boldsymbol{\ell}_k^*}_{\text{held wrist changes, tracker errors and branch choice}} \\ & + \underbrace{\left[\Delta\theta_{k,j}^* - (\theta_{k,j} - \theta_{k,0})\right]\boldsymbol{\ell}_k^*}_{\text{physical travel differs from encoder change}}. \end{aligned}$$

This decomposition does not assert that either contribution vanishes. The travel discrepancy itself is

$$\begin{aligned} \Delta\theta_{k,j}^* - (\theta_{k,j} - \theta_{k,0}) = & \Delta\theta_{k,\text{sys}}^{\text{rob}}(\theta_{k,j}) - \Delta\theta_{k,\text{sys}}^{\text{rob}}(\theta_{k,0}) \\ & + \Delta\theta_{k,\text{rnd},j}^{\text{rob}} - \Delta\theta_{k,\text{rnd},0}^{\text{rob}}. \end{aligned}$$

With exactly collinear rotation vectors,

$$\sum_j w_j(\theta_{k,j} - \theta_{k,0})\boldsymbol{v}_j = \boldsymbol{\ell}_k^* \sum_j w_j(\theta_{k,j} - \theta_{k,0})\Delta\theta_{k,j}^*.$$

Thus changing travel magnitudes alone leaves the normalized direction unchanged when the scalar sum is positive. With observation errors it changes the sample contributions.

The rotation-vector formula divides by $\sin\beta_j$. The implementation has a near-zero approximation but no dedicated near-π case; the direction-only sample filter does not prove stability at π or guarantee the correct physical angle branch. Without other errors, a wrong branch gives

$$\Delta\theta_{k,j,\text{est}} - \Delta\theta_{k,j}^* = 2\pi m, \qquad m \in \mathbb{Z} \setminus \{0\}.$$

This is not a small-angle error.

Step B. The position-difference residual at the true stage direction separates as

$$\begin{aligned}
&\boldsymbol{p}_{m,\mathrm{obs},j} - \boldsymbol{p}_{m,\mathrm{obs},0} - (\theta_{k,j} - \theta_{k,0})\boldsymbol{\ell}_k^* \\
&\quad = \underbrace{\boldsymbol{p}_{m,j}^* - \boldsymbol{p}_{m,0}^* - \Delta\theta_{k,j}^*\boldsymbol{\ell}_k^*}_{\text{held-joint changes}} \\
&\qquad + \underbrace{(\boldsymbol{p}_{m,\mathrm{obs},j} - \boldsymbol{p}_{m,j}^*) - (\boldsymbol{p}_{m,\mathrm{obs},0} - \boldsymbol{p}_{m,0}^*)}_{\text{tracker errors}} \\
&\qquad + \left[\Delta\theta_{k,j}^* - (\theta_{k,j} - \theta_{k,0})\right]\boldsymbol{\ell}_k^*.
\end{aligned}$$

Using position differences avoids multiplying the reference position by an observed relative rotation. It does not remove actual held-wrist changes, which can move the tool origin.

For the projected stage travel, adding and subtracting the physical single-axis displacement gives

$$\begin{aligned}
\Delta\theta_{k,j,\mathrm{est}} - \Delta\theta_{k,j}^* = \Delta\theta_{k,j}^*(\boldsymbol{\ell}_{k,\mathrm{est}}^T\boldsymbol{\ell}_k^* - 1) \\
+ \boldsymbol{\ell}_{k,\mathrm{est}}^T\left[\boldsymbol{p}_{m,\mathrm{obs},j} - \boldsymbol{p}_{m,\mathrm{obs},0} - \Delta\theta_{k,j}^*\boldsymbol{\ell}_k^*\right].
\end{aligned}$$

For both stages and wrists, distortion initialization then has the exact error

$$\begin{aligned}
y_{k,j} - \Delta\theta_{k,\mathrm{sys}}^{\mathrm{rob}}(\theta_{k,j}) = (\theta_{k,0,\mathrm{est}} - \theta_{k,0}^*) + (\Delta\theta_{k,j,\mathrm{est}} - \Delta\theta_{k,j}^*) \\
+ \Delta\theta_{k,\mathrm{rnd},j}^{\mathrm{rob}}.
\end{aligned}$$

The joint fit replaces the encoder-travel approximation by the estimated distortion function, but it does not recover each random joint realization.

3.4.5 Wrist-center, tool and base initialization errors

The center fit uses $\boldsymbol{p}_{m,\mathrm{obs},j} - R_{m,\mathrm{obs},j}\boldsymbol{s} - \boldsymbol{o}$. At the true tool vector $\boldsymbol{s}$ and reference center $\boldsymbol{o}_w$,

$$\begin{aligned}
&\boldsymbol{p}_{m,\mathrm{obs},j} - R_{m,\mathrm{obs},j}\boldsymbol{s} - \boldsymbol{o}_w \\
&\quad = \underbrace{R_X^*\sum_{i=1}^{3}(\theta_{i,j}^* - \theta_{i,0}^*)\boldsymbol{d}_i^*}_{\text{stage-induced wrist-center motion}} \\
&\qquad + \underbrace{(\boldsymbol{p}_{m,\mathrm{obs},j} - \boldsymbol{p}_{m,j}^*) - (R_{m,\mathrm{obs},j} - R_{m,j}^*)\boldsymbol{s}}_{\text{tracker pose errors}}.
\end{aligned}$$

Only stage changes move $\boldsymbol{o}_w$ in this spherical-wrist model. The center-fit coefficient matrix also contains uncertain $R_{m,\mathrm{obs},j}$; ordinary least squares treats that matrix as fixed.

The true and initialized base positions satisfy

$$\begin{aligned}
\boldsymbol{p}_X^* &= \boldsymbol{o}_w - R_X^*\left(\sum_{i=1}^{3}\theta_{i,0}^*\boldsymbol{d}_i^* + \boldsymbol{p}_{\mathrm{cw}}\right), \\
\boldsymbol{p}_{X,\mathrm{est}} &= \boldsymbol{o}_{w,\mathrm{est}} - R_{X,\mathrm{est}}\left(\sum_{i=1}^{3}\theta_{i,0,\mathrm{est}}\boldsymbol{d}_{i,\mathrm{est}} + \boldsymbol{p}_{\mathrm{cw}}\right).
\end{aligned}$$

These equations retain center, rotation, reference-joint and internal-axis errors. The initial tool rotation additionally uses the shared reference rotation; tool translation uses the stacked center fit.

3.4.6 Averaging independent arrivals and retained errors

Joint and tracker random errors. Let $C_{\text{joint},j}$ and $C_{\text{tracker},j}$ be their first-order pose covariances at one arrival. For n_j independent arrivals,

$$C_{\text{mean},j} \approx \frac{C_{\text{joint},j} + C_{\text{tracker},j}}{n_j} + C_{\text{shared},j}.$$

Thus revisits reduce both independent contributions. Shared-reference errors remain in $C_{\text{shared},j}$; fixed tracker distortion and systematic joint distortion are biases, not terms divided by n_j.

Effect of pose averaging on the objective. For one reading per arrival, all observations at a command receive equal weight. Position averaging preserves the quadratic position objective up to a parameter-independent term:

$$\sum_{q=1}^{n_j} \left\| \boldsymbol{p}_{m,\text{obs},j}^{(q)} - \hat{\boldsymbol{p}}_{m,j} \right\|_2^2 = n_j \left\| \boldsymbol{p}_{m,\text{obs},j} - \hat{\boldsymbol{p}}_{m,j} \right\|_2^2 + \underbrace{\sum_{q=1}^{n_j} \left\| \boldsymbol{p}_{m,\text{obs},j}^{(q)} - \boldsymbol{p}_{m,\text{obs},j} \right\|_2^2}_{\text{independent of the fitted parameters}}.$$

For the rotation-vector residual $\boldsymbol{\alpha}_j^{(q)}(\boldsymbol{x})$ of an individual arrival, the corresponding identity does not hold in general:

$$\sum_{q=1}^{n_j} \left\| \boldsymbol{\alpha}_j^{(q)}(\boldsymbol{x}) \right\|_2^2 \not\equiv n_j \left\| \boldsymbol{\alpha}_j(\boldsymbol{x}) \right\|_2^2 + \text{constant}.$$

The rotation average is exact for its coordinate-axis least-squares objective, but combining it with rotation-vector residuals in the joint fit is an approximation.

Visit-count weights. With $w_j = n_j$, the adopted residual weighting assumes

$$C_{\text{assumed},j} = \frac{1}{n_j} \begin{bmatrix} \sigma_R^2 I & 0 \\ 0 & \sigma_p^2 I \end{bmatrix}.$$

The count reflects independent arrivals, but the two fixed scales still omit pose-dependent covariances, rotation–translation correlations and shared-reference uncertainty.

3.4.7 The base turn that tracker observations cannot determine

The invariance $F_X \hat{F}_{\text{ee}} = (F_X Q(c))(Q(-c)\hat{F}_{\text{ee}})$ must be compatible with the robot chain, not merely with matrix multiplication. A rotation $Q(-c)$ about $\boldsymbol{e}_x$ commutes with the known translations F_{cw} and F_{we}, both of which lie along $\boldsymbol{e}_x$. Moving it through each joint transform rotates the joint axis and leaves its angle unchanged:

$$\begin{aligned} Q(-c)\,\text{Frame}(I, \boldsymbol{p}) &= \text{Frame}(I, R(\boldsymbol{e}_x, -c)\boldsymbol{p})Q(-c), \\ Q(-c)\,\text{Frame}(R(\boldsymbol{d}, \theta), \boldsymbol{0}) &= \text{Frame}(R(R(\boldsymbol{e}_x, -c)\boldsymbol{d}, \theta), \boldsymbol{0})Q(-c). \end{aligned}$$

Applying these identities along the complete chain gives another model with

$$\boldsymbol{d}_k' = R(\boldsymbol{e}_x, -c)\boldsymbol{d}_{k,\text{est}}, \qquad \Delta F_{\text{we}}' = Q(-c)\Delta F_{\text{we,est}}, \qquad F_X' = F_{X,\text{est}} Q(c).$$

The transformed model satisfies

$$\boldsymbol{d}'_1 = \boldsymbol{e}_x, \qquad \theta'_{k,j} = \theta_{k,j,\mathrm{est}}, \qquad F'_X \hat{F}'_{\mathrm{ee}} = F_{X,\mathrm{est}} \hat{F}_{\mathrm{ee}}, \qquad (\boldsymbol{d}'_i)^T \boldsymbol{d}'_k = \boldsymbol{d}_{i,\mathrm{est}}^T \boldsymbol{d}_{k,\mathrm{est}}.$$

Neither tracker observations nor homing distinguishes c. Bounds restrict it, but the noisy finite-grid accepted set need not contain the true turn; the fallback can still violate a bound.

The two constraints serve different purposes:

$\boldsymbol{\phi}_2^T \boldsymbol{e}_x = 0$	fixes a coordinate representative,
$\Delta\theta^{\mathrm{rob}}_{k,\mathrm{sys,est}}(\bar{\theta}_{\mathrm{encoders},k}) = -\bar{\theta}_{\mathrm{encoders},k}$	constrains constant distortion shifts.

Neither removes finite-data uncertainty or proves full rank for a particular observation set.

For the numerical rank check, use central differences with increments $10^{-4}\,\mathrm{mm}$ or $10^{-6}\,\mathrm{rad}$. With singular values $s_a(J)$, the implementation reports

$$\mathrm{rank}_{\mathrm{num}}\, J = \#\{a : s_a(J) > 10^{-8} s_{\max}(J)\}.$$

Small singular values indicate weakly constrained parameter combinations. Inferred standard deviations remain conditional on the adopted weighting and local model.

In particular,

$$\hat{F}'_{\mathrm{ee}} = Q(-c)\hat{F}_{\mathrm{ee}} \neq \hat{F}_{\mathrm{ee}} \quad \text{while} \quad F'_X \hat{F}'_{\mathrm{ee}} = F_X \hat{F}_{\mathrm{ee}}.$$

A small tracker residual alone cannot certify base-relative forward accuracy.

3.4.8 What joint fitting can improve and what remains

The pose fit refines fixed geometry and selected distortion values. Random move realizations and tracker systematic distortion remain unmodeled; their effects can remain in residuals or be absorbed into fitted parameters.

If the free-parameter Jacobian has full column rank, its local sensitivity satisfies

$$\boldsymbol{\delta} \approx -J^{\dagger} \boldsymbol{r}(\boldsymbol{x}_{\mathrm{true}}), \qquad \|\boldsymbol{\delta}\|_2 \lesssim \frac{\|\boldsymbol{r}(\boldsymbol{x}_{\mathrm{true}})\|_2}{s_{\min}(J)}.$$

Here $\boldsymbol{x}_{\mathrm{true}}$ must be representable in the chosen model, including its fixed home values; $s_{\min}$ is the smallest singular value. The relation is local and depends on parameter units.

For selected-position spacing h_k, the noiseless piecewise-linear approximation has the conservative Lipschitz bound $\omega^{\mathrm{rob}}_{k,\mathrm{sys}} h_k/2$. With uncertain fitted endpoint values, a corresponding interval error bound includes

$$\max\big(\text{absolute error at either endpoint}\big) + \frac{1}{2}\omega^{\mathrm{rob}}_{k,\mathrm{sys}} h_k.$$

After relocating the home selected position, use

$$h_k = \max_l \big(\theta_k^{(l+1)} - \theta_k^{(l)}\big).$$

Smaller h_k reduces the interpolation term, not endpoint uncertainty. The bound does not cover extrapolation; additional free values require additional independent constraints.

Verify final forward and inverse errors on independent workspace configurations. Ground truth is used only for validation and parameter comparison.

4 Algorithmic Approach

The implementation uses Python 3.11 with NumPy for linear algebra and `scipy.optimize.least_squares` for the non-linear fit; the unit tests use `pytest`. The course-supplied `src` and `data_types` modules provide the simulator and the vector, rotation and frame classes. The pipeline follows Method 1 of the handout: home the robot, joint collection each joint through its selected encoder positions, construct parameter initial values for Eq. (6), refine them by weighted least squares, and evaluate ImprovedFwdKins and ImprovedInvKins over the work volume. Ground truth is used only in the evaluation (Steps 8–10).

4.1 Scene design (Step 1)

Main tasks. Step 1 fixes the robot base, the tracker pose and the layout of $mBee$.

i. The robot base is placed at the workspace origin, $F_{\text{rob}} = \text{Frame}(I, \mathbf{0})$. A point $\boldsymbol{p}$ of the work volume moves by at most $\epsilon_{\text{rob}}^{\max} + \alpha_{\text{rob}}^{\max} \|\boldsymbol{p}\|_2$ under the base error, so a work volume near the origin keeps this shift small.

ii. The tracker views the work volume from the side, about 1.7 m from its center $(840, 0, 200)$ mm. The pose was chosen by a grid search over viewing directions and distances so that every calibration stop of both plans stays inside the viewing volume with a clearance larger than that worst-case shift, which `plan_calibration_poses` checks for every stop. All 729 test poses of Step 9 are visible.

iii. $mBee$ is a regular tetrahedron of radius $r = 74.9$ mm, just inside $\|\boldsymbol{a}_k^*\|_2 \leq 75$ mm, centred on the origin of $\{e\}$. The layout and the per-component reading-noise scales $\sigma_R = 8.2 \times 10^{-4}$ rad and $\sigma_p = 0.05$ mm are derived in Section 3.3.8.

iv. No other marker body is defined. The tracker does not jiggle, so its frame is a fixed reference and needs no reference body, and $mBee$ needs no refinement because its layout is exact.

4.2 Homing (Step 6)

Main task. `measure_home_offsets` calls HomeRobot $H = 5$ times and returns

$$\Delta\theta_{k,\text{sys,est}}^{\text{rob}}(\bar{\theta}_{\text{encoders},k}) = -\bar{\theta}_{\text{encoders},k}, \tag{27}$$

as in Eq. (15). Each homing after the first costs about t_{settle}. The values fix one selected position per joint in Section 4.6. Step 6 then reads ten diagnostic poses to show the error of the nominal kinematics; these readings are not used by the calibration.

4.3 Pose planning (Step 7)

Main tasks. Step 7 chooses the selected encoder positions, the number of readings per stop and the order of the stops, and checks every stop against the joint limits and the viewing volume.

Table 11: Planning functions (Step 7).

Step	Function	Inputs	Outputs
7	`joint_knots`	Joint limits; $h_{\text{stage}} = 100\,\text{mm}$, $h_{\text{wrist}} = \pi/4$.	Selected encoder positions $\theta_k^{(l)}$ of each joint.
7	`readings_per_stop`	Marker layout; targets 10^{-3} rad and 0.2 mm.	Repetitions n_{stage}, n_{wrist}.
7	`plan_calibration_poses`	Robot, tracker, layout, target, plan options.	Ordered joint commands.
7	`estimate_plan_time`	Robot, commands.	Simulated time of the plan.

i. `joint_knots` spaces the selected encoder positions of each joint evenly over its full range, both ends included, at most h_k apart: 7 selected encoder positions for each stage (100 mm), 7 for joints 4 and 5, and 9 for joint 6 ($\pi/4$). The bound $\frac{1}{2}\omega_{k,\text{sys}}^{\text{rob}} h_k$ gives interpolation errors of at most 0.025 mm per stage and 2.0×10^{-5} rad per wrist joint.

ii. `readings_per_stop` sizes the repetitions of each stop. A stop listed n times in a row is re-commanded and re-read, which costs $t_{\text{settle}} + \Delta t_{\text{samp}}$ and averages both the reading noise and the joint repeatability. With $\sigma_{\text{wrist}}^2 = \sigma_R^2 + (\sigma_{k,\text{rnd}}^{\text{rob}})^2$ and $\sigma_{\text{stage}}^2 = \sigma_p^2 + (\sigma_{k,\text{rnd}}^{\text{rob}})^2$, the implementation uses the independent-sample standard-error rule $\sigma/\sqrt{n} \le f \cdot \text{target}$ to choose

$$n = \left\lceil \frac{\sigma^2}{(f \cdot \text{target})^2} \right\rceil. \tag{28}$$

With the implementation choice $f = 0.1$, this gives $n_{\text{stage}} = 7$ and $n_{\text{wrist}} = 68$. The imported implementation sends these repeated commands consecutively. The separated-arrival requirement of Section 3.3.1 is not yet implemented; the saved times and results below describe the consecutive-command plan.

iii. `plan_calibration_poses` with `method="single_axis"` implements hint 1(b): each joint in turn is swept through all its selected encoder positions, starting at the nearer end of its range, and the robot returns to the reference command after each axis. The reference command is the encoder configuration $\boldsymbol{\theta} = \mathbf{0}$. The home encoder reading can lie outside the command limits of joints 1 and 3, so it is not used as the reference command. The home distortion constraint is applied separately in the model. The plan has 39 distinct poses and 1716 readings.

iv. Every stop must keep all markers inside the viewing volume with a clearance of $\epsilon_{\text{rob}}^{\max} + \alpha_{\text{rob}}^{\max} \|\boldsymbol{p}\|_2$, evaluated with the nominal F_{rob} and F_{ee}. The planner raises an error rather than dropping a stop, since a missing stop would leave a selected position interval without data.

v. `estimate_plan_time` sums the simulator motion time and tracker sampling time for every command. The single-axis plan takes about 530–545 s, depending on the starting configuration; travel accounts for about 340 s, and the stops and repetitions for the rest.

vi. `method="concurrent"` is a faster variant used for comparison: the wrist joints 4, 5, 6 are swept together with the stages 1, 2, 3, so the stage travel shares the wrist's motion time. Its data no longer consist of single-axis joint collections, so the decoupled estimate of Section 4.5 does not apply.

4.4 Data collection and nominal base estimate (Steps 6–7)

Main tasks. `collect_calibration_data` moves the robot to each command with MoveJoints and reads *mBee* with SampleMarkerBodies. The `repeats` readings at one command are averaged with `average_frames`; they share the joint repeatability error, so this averaging removes only the reading noise. A pose at which *mBee* is not visible is recorded as skipped after one reading, whose time is still charged. `estimate_base_frame` then gives the initial base

$$F_{X,\text{init}} = \text{avg}_j\big(F_{m,\text{obs},j}\, F_{\text{ee}}(\boldsymbol{\theta}_j)^{-1}\big) \tag{29}$$

with the nominal kinematics, where avg uses the position mean and rotation least squares of `average_frames`. The notebook uses `repeats=1`; repeated commands in the plan therefore produce new moves, whereas `repeats>1` would read repeatedly without another move.

4.5 Decoupled initial estimate (Step 8)

Main task. `initial_estimate` uses the single-joint observations to construct the initial values of Sections 3.3.2–3.3.6:

i. *Stages.* Fit tracker-frame directions from position differences and encoder changes, then project the differences onto the fitted directions to estimate relative physical travel.

ii. *Wrist.* Convert relative rotations to rotation vectors, fit tracker-frame axes using encoder changes, and unwrap the signed angles to estimate relative physical travel. The 2.8 rad filter applies only to axis-direction fitting.

iii. *Reference values and distortions.* Use the home offsets to initialize reference physical joint values, add relative travel, subtract commands, and interpolate the resulting distortions onto the selected encoder positions. Fix the measured home distortion of each joint.

iv. *Base rotation and axes.* Construct $R_{X,\text{est}}$ from the estimated stage directions under the coordinate convention of Section 3.3.5, then remove the reference wrist rotations to estimate the internal axes and tilts.

v. *Tool and translation.* Estimate the tool rotation from the reference pose. Jointly fit $\boldsymbol{s}_{\text{est}}$ and $\boldsymbol{o}_{w,\text{est}}$ using Eq. (17), then recover tool and base translations.

These are initial estimates from noisy observations. Zero observed direction excitation, parallel stage directions and a rank-deficient center system are rejected before initialization is returned. On noise-free data this estimate is within 10^{-4} rad and 0.01 mm of the generating model. On simulated data (25 seeds) its RMS residuals are $(3.4\text{–}4.6) \times 10^{-4}$ rad and 0.017–0.026 mm, against 0.014–0.023 rad and 2.5–4.9 mm for the nominal geometry, and already close to those after the fit.

4.6 Weighted least-squares fit (Step 8)

Main tasks. `fit` refines the decoupled estimate over the full model Eq. (6).

i. *Parameters.* $\boldsymbol{x}$ holds the base correction relative to $F_{X,\text{init}}$ (6), the tilts (9: two for each of $\boldsymbol{d}_3, \boldsymbol{d}_4, \boldsymbol{d}_5, \boldsymbol{d}_6$ and the z-component for $\boldsymbol{d}_2$), the tool correction parameters (6) and the selected distortion values other than the fixed home values (38), 59 in all. $\Delta\theta^{\text{rob}}_{k,\text{sys,est}}$ is piecewise linear in the selected distortion values, Eq. (7), and extrapolated linearly beyond the end selected encoder positions.

ii. *Residual.* For sample j, obtain $\boldsymbol{\alpha}_j, \boldsymbol{\epsilon}_j$ from $(F_X(\boldsymbol{x})\hat{F}_{\text{ee}}(\boldsymbol{\theta}_j;\boldsymbol{x}))^{-1}F_{m,\text{obs},j}$, as in Eq. (18). The rotation and translation parts are divided by σ_R and σ_p. These scales describe marker random noise; the residual covariance also contains joint and systematic errors, as discussed in Section 3.4.6.

iii. *Merging.* Samples with the same command are averaged and weighted by $\sqrt{n}$, giving 39 instead of 1716 residual blocks. Exact equivalence to fitting all individual poses is not guaranteed for the rotation residual (Section 3.4.6).

iv. *Solver.* `least_squares` (trust-region reflective, Jacobian scaling, tolerances 10^{-12}) starts from the decoupled estimate and converges in 9–19 evaluations over 25 seeds.

v. *Rank.* `jacobian_rank_report` differentiates the weighted residuals by central differences and reports the singular values; their inverses describe local sensitivity along the right singular vectors. Interpreting them as standard deviations additionally requires a full-rank linearization and correctly whitened independent residuals.

4.7 Rotation about $\boldsymbol{d}_1$ (Step 8)

Section 3.4.7 derives the remaining freedom to turn the base about $\boldsymbol{e}_x$ while oppositely turning the internal axes and tool. The fit fixes $\boldsymbol{\phi}_2^T\boldsymbol{e}_x = 0$ to choose coordinates; this condition alone does not guarantee a full-rank Jacobian for arbitrary data. Afterwards `choose_gauge` applies the transformation of Section 3.3.9, preserving every tracker-frame prediction. It selects the midpoint between the extreme grid points satisfying the axis-tilt, tool-rotation and base-placement bounds, or the least-violating point if none is accepted. The base condition uses the specified tracker pose and nominal F_{rob}. The selected turn is not a measurement of the true turn.

4.8 Improved forward and inverse kinematics

`fwd_kins` evaluates Eq. (6) with $\theta_{k,\text{est}} = \theta_k + \Delta\theta^{\text{rob}}_{k,\text{sys,est}}(\theta_k)$, $\boldsymbol{d}_{k,\text{est}} = R(\boldsymbol{\phi}_{k,\text{est}})\boldsymbol{d}_k$ and $F_{\text{we}}\Delta F_{\text{we,est}}$, followed by left multiplication by $Q(-c)$, as in Section 3.3.10. `inv_kins`, based on the template solver, seeds both wrist branches from the nominal closed form, refines them by Levenberg–Marquardt steps on the pose error of `fwd_kins`, Section 3.3.10, and returns the converged in-limit solution nearest the seed. The residual is checked after any angle wrapping; iteration-limit exits that fail the tolerance raise an error.

4.9 Evaluation (Steps 9–10)

Main tasks. Step 9 evaluates ImprovedFwdKins on the fixed grid of the template (729 poses, all visible): for each $\boldsymbol{\theta}_i$,

$$\Delta F_i = \left(F^{-1}_{\text{tracker}}F^{*}_{\text{rob}}F^{*}_{\text{ee}}(\boldsymbol{\theta}_i)\right)^{-1}\left(F_{X,\text{est}}Q(c)\right)\text{ImprovedFwdKins}(\boldsymbol{\theta}_i). \tag{30}$$

Step 10 takes 40 grid poses as targets F in the tracker frame, commands

$$\boldsymbol{\theta} = \text{ImprovedInvKins}\left(\left(F_{X,\text{est}}Q(c)\right)^{-1}F\right) \tag{31}$$

with MoveJoints, and compares F with the pose actually reached. Both metrics are expressed in the tracker frame, as in the template: there the base estimate and the kinematics enter together, so the unobservable rotation c cancels. The handout's definition relative to the robot base is reported separately in Section 7. The targets are visited in a greedy nearest-first order of motion time, which leaves the targets unchanged and shortens the testing time.

4.10 Function-level execution flow

Figure 2 shows the call sequence of Steps 6–10.

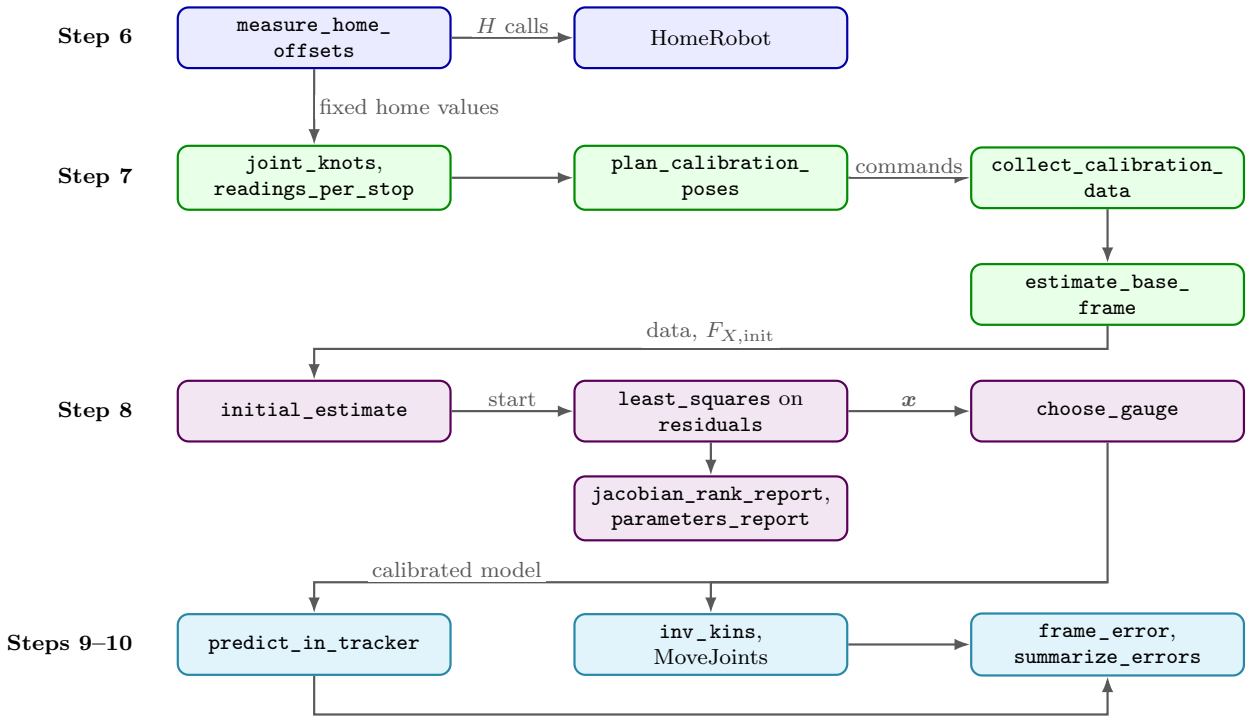


Figure 2: Execution flow (Steps 6–10).

5 Program Structure

The program consists of four student modules under `src/student/`, one notebook that drives the calibration and its evaluation, and two student `pytest` test files. The simulator in `src/simulation/` and the nominal robot in `src/robot.py` are supplied by the course and are not modified.

5.1 File and module organization

Table 12: Program files and modules.

Location	Responsibility
`notebooks/assignment2_template.ipynb`	Defines the scene (Step 1), builds the simulation, runs the unit tests, homes the robot, plans and collects the calibration data, fits the improved kinematics, evaluates them (Steps 9–10), compares plans over several seeds, and runs the extended evaluation (25 seeds; full joint ranges and the base-frame definition).
`src/student/metrics.py`	Pose-error metric $\Delta F = F_a^{-1} F_b$ and error statistics.
`src/student/registration.py`	Point-set registration, frame averaging and optional marker-body refinement.

Location	Responsibility
`src/student/workflow.py`	Homing, pose planning, data collection, simulated-time estimate and the nominal base estimate.
`src/student/kinematics.py`	`ImprovedKinematics`: error-model parameterization, ImprovedFwdKins, ImprovedInvKins, decoupled initial estimate, least-squares fit, choice of the rotation about $\boldsymbol{d}_1$, Jacobian rank report and parameter report.
`tests/test_student.py`	Unit tests of the four student modules (Section 6).
`tests/test_kinematics_validation.py`	Independent forward-kinematics answers, interpolation/home constraints, inverse edge cases and unobservable-data rejection.
`tests/kinematics_sensitivity.py`	Controlled noise, arrival-count and spacing experiments; saves per-run CSV data and the report figure.
`tests/test_simulation.py`	Course-supplied tests of the simulator.
`src/robot.py`, `src/simulation/`	Course-supplied nominal robot, simulator, simulated clock and constants (not modified).
`data_types/`, `assignments/assignment1/src/`	Course-supplied vector, rotation and frame types, tracker and marker-body classes.

5.2 Program hierarchy

Figure 3 shows the dependencies between the notebook and the modules. Each arrow points from a caller to the module it uses.

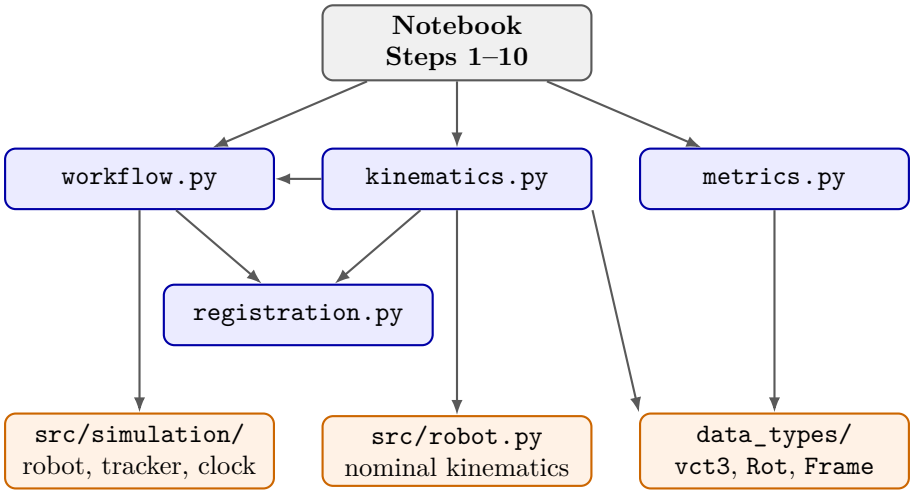


Figure 3: Program hierarchy. Blue: student modules; orange: course-supplied modules.

5.3 Runtime functions

Table 13 lists the public functions of the student modules and the notebook step that first calls each. ImprovedInvKins (`ImprovedKinematics.inv_kins`) is based on the template solver, checks final convergence, and iterates on our `fwd_kins`.

Table 13: Runtime functions.

Step	Function	Inputs	Output and purpose
4	`metrics.frame_error(F_a,F_b)`	Reference and compared frames.	$(\|\boldsymbol{\alpha}\|_2, \|\boldsymbol{\epsilon}\|_2)$ of $F_a^{-1}F_b$; every accuracy number uses it.
4	`metrics.summarize_errors(errors)`	List of error pairs.	Count, RMS, mean and maximum of angle and position.
5	`registration.register_points(a,b)`	Corresponding point lists.	Least-squares rigid frame F with $F\boldsymbol{a}_i \approx \boldsymbol{b}_i$; rejects fewer than three pairs and collinear sets.
5	`registration.average_frames(frames)`	Estimates of one frame.	Chordal mean rotation projected onto $SO(3)$ and mean translation.
5	`registration.refine_marker_body(trk,placed,n)`	Tracker, placed body, number of readings.	Averaged body-frame marker layout, stored with UpdateMarkerBody (optional; not used by the calibration).
6	`workflow.measure_home_offsets(rob,n)`	Robot, number of homings.	$\Delta\theta^{\text{rob}}_{k,\text{sys,est}}(\bar{\theta}_{\text{encoders},k}) = -\bar{\theta}_{\text{encoders},k}$.
6, 7	`workflow.collect_calibration_data(rob,trk,body,thetas,repeats)`	Robot, tracker, calibration body, commands.	`CalibrationData`: commands, observed frames $F_{m,\text{obs},j}$, readings per sample, skipped poses, collection time.
6, 7	`workflow.estimate_base_frame(data,kins)`	Data and a kinematic model.	Average of $F_{m,\text{obs},j}F_{\text{ee}}(\boldsymbol{\theta}_j)^{-1}$, i.e. $F_{X,\text{init}}$ with the nominal model.
7	`workflow.joint_knots(rob,h_stage,h_wrist)`	Robot, knot spacings.	Knot positions $\theta_k^{(l)}$ over each joint range.
7	`workflow.readings_per_stop(layout,acc,...)`	Marker layout, angular and position targets.	Repetitions per stage stop and per wrist stop.
7	`workflow.plan_calibration_poses(rob,trk,layout,acc,...)`	Robot, tracker, layout, target; plan options.	Ordered joint commands; raises if a stop is outside the limits or too close to the edge of the viewing volume.
7	`workflow.estimate_plan_time(rob,thetas)`	Robot, commands.	Simulated time of the plan (motion plus readings).
8	`ImprovedKinematics(rob,F_X_init,joint_offsets)`	Robot, initial base, home offsets.	Model with knots, home knots and zero parameters.
8	`ImprovedKinematics.fit(data,F_tracker)`	Calibration data, specified tracker pose.	Decoupled start, weighted least squares, choice of the rotation about $\boldsymbol{d}_1$; returns residuals, iterations and the chosen gauge.
8	`ImprovedKinematics.initial_estimate(data)`	Single-axis sweep data.	Closed-form parameter estimate (Section 4.5).
8	`ImprovedKinematics.choose_gauge(F_tracker)`	Specified tracker pose.	Rotation c about $\boldsymbol{d}_1$ and its admissible interval.
8	`ImprovedKinematics.jacobian_rank_report(data)`	Calibration data.	Singular values, rank, condition number and parameter standard deviations.
8	`ImprovedKinematics.parameters_report()`	None.	Axes, axis angles, tilts, knot values, joint offsets and scales, tool twist, gauge.
9	`ImprovedKinematics.fwd_kins(theta)`, `predict_in_tracker(theta)`	Encoder command.	ImprovedFwdKins($\boldsymbol{\theta}$) relative to the base, and $\big(F_{X,\text{est}}Q(c)\big)$ ImprovedFwdKins($\boldsymbol{\theta}$).
10	`ImprovedKinematics.inv_kins(F,seed)`	Pose relative to the base, seed command.	ImprovedInvKins(F): converged in-limit command nearest the seed; raises if no candidate passes the final residual check.

5.4 Tests

`tests/test_simulation.py` contains 23 course-supplied simulator cases. The original `tests/test_student.py` has 23 cases covering metrics, registration, workflow and fitting. The additional `tests/test_kinematics_validation.py` has 35 cases, for 81 in total. Section 6 describes their inputs, reference answers and assertions. The controlled sensitivity experiments are separate from these pass/fail tests.

5.5 Program outputs

The notebook reports

- **Calibration:** home offsets, plan size and simulated collection time, fit residuals, Jacobian rank, the chosen rotation about $\boldsymbol{d}_1$ and the estimated parameters with their ground-truth comparison;
- **Evaluation:** forward and inverse errors of the nominal and improved kinematics, and the simulated testing time;
- **Several seeds:** calibration time, testing time and maximum errors for both plans and five seeds;
- **Extended evaluation:** worst errors and fit statistics over seeds 1–25, and, for seeds 1–8, forward and inverse errors on random full-range configurations in the tracker frame and relative to the robot base.

The saved original notebook run took about 4 minutes. The additional controlled experiments are run separately with `pythontests/kinematics_sensitivity.py`; their data are stored in `tests/data/` and their plot in `report/figures/`.

6 Validation Approach

Validation has two parts. End-to-end tests measure the accuracy of the calibrated kinematics over the work volume against the hidden ground truth (Steps 9–10). Unit tests check each component in isolation, mostly on synthetic data with a known answer. Hidden simulator values are used only to score completed estimates.

6.1 End-to-end accuracy

For each evaluated configuration the error is $\Delta F = \mathrm{Frame}(R(\boldsymbol{\alpha}), \boldsymbol{\epsilon})$, with $0 \le \|\boldsymbol{\alpha}\|_2 \le \pi$ (`metrics.frame_error`). The forward error is Eq. (30) on the 729 poses of the template's grid, $\theta_{1,3} \in \{100, 300, 500\}$ mm, $\theta_2 \in \{-200, 0, 200\}$ mm, $\theta_{4,5} \in \{-1.3, 0, 1.3\}$ and $\theta_6 \in \{-2, 0, 2\}$ rad. The inverse error compares 40 targets drawn from this grid with the poses reached after MoveJoints, so it includes the repeatability of the final move. Both are reported as RMS and maximum over the poses and compared with the handout's targets, 10^{-3} rad and 0.2 mm, applied to the maximum. The complete pipeline is repeated for five robot and tracker seeds and for both plans, and for 20 further seeds with the single-axis plan. The notebook's extended evaluation also takes 600 random configurations over the full joint ranges for seeds 1–8 and evaluates the handout's definition relative to the robot base, $(F^*_{\mathrm{ee}}(\boldsymbol{\theta}))^{-1}\,\mathrm{ImprovedFwdKins}(\boldsymbol{\theta})$, together with an inverse error for targets specified relative to the base (Section 7.5).

6.2 Unit tests

`tests/test_simulation.py` (23 tests, supplied) checks the simulator: the error bounds, homing, motion time, joint limits, nominal kinematics, viewing volume, clock and calibration-body rules. `tests/test_student.py` checks the student modules. Tables 14 and 15 list its 23 cases. Their fitting data are generated by `ImprovedKinematics` with known parameters, so those tests check recovery and internal consistency. Independent forward answers and inverse edge cases are added below.

Table 14: Supplied unit tests of the student modules.

Module	Test	Purpose
A	`frame_error_known_values`	Known rotation and translation errors are reproduced.
A	`frame_error_is_invariant_to_common_left_factor`	The metric does not depend on a common reference frame.
A	`summarize_errors`	Count, RMS and maximum are correct.
B	`register_points_recovers_known_transform`	Noise-free registration recovers the transform to 10^{-9}.
B	`register_points_rejects_degenerate_input`	Fewer than three or collinear points are rejected.
B	`average_frames_of_identical_frames`	The average of identical frames is that frame.
B	`average_frames_is_a_proper_rotation`	The averaged rotation is orthonormal with determinant $+1$.
C	`estimate_base_frame_on_synthetic_data`	The base is recovered exactly from nominal data.
C	`measure_home_offsets_matches_encoders`	The home offsets agree with the encoder reading at home within four times the largest $\sigma_{k,\mathrm{rnd}}$.
D	`improved_kinematics_initial_params_match_nominal`	Zero parameters reproduce the nominal kinematics.
D	`improved_inv_kins_round_trip`	ImprovedInvKins inverts ImprovedFwdKins to 10^{-6} mm and 10^{-8} rad.

Table 15: Unit tests written for this design (A–D: metrics, registration, workflow, kinematics; †: TODO test of the template; ×2: run for both plans).

Module	Test	Purpose
B	`refine_marker_body_reduces_layout_error`†	Averaging 50 readings reduces the error of a layout with manufacturing error to less than 30% of its initial value, at a cost of exactly $50\,\Delta t_{\mathrm{samp}}$.
C	`planned_poses_are_visible_and_within_limits`† (×2)	Every stop is within the joint limits and visible, every knot is visited, and each stop is repeated the planned number of times.
C	`plan_rejects_a_tracker_that_cannot_see_the_sweeps`	A tracker that cannot see the sweeps raises an error instead of silently dropping stops.
C	`readings_per_stop_grows_with_accuracy`	Halving both targets at least triples the repetitions, as Eq. (28) requires.

Module	Test	Purpose
C	`collect_calibration_data_averages_readings_and_skips_invisible_poses`	Repeated readings form one sample close to the actual pose; an invisible pose is skipped after one reading with a warning; the collection time equals motion plus readings.
C	`estimate_plan_time_matches_the_simulated_clock`	The planned time equals the clock advance of collecting the plan.
D	`fit_recovers_synthetic_parameters`† (×2)	From a base off by 0.05 rad and 5 mm, the fit reproduces the generating model to 10^{-8} rad and 10^{-5} mm, checks the fitted axis tilts, and verifies full numerical Jacobian rank for these data.
D	`decoupled_initial_estimate_is_close_to_the_truth`	The closed-form estimate alone is within 10^{-4} rad and 0.01 mm of the generating model.
D	`decoupled_initial_estimate_needs_single_axis_sweeps`	Data without single-axis sweeps are rejected.
D	`gauge_choice_keeps_tracker_poses_and_respects_the_bounds`	The choice of c leaves all tracker-frame poses unchanged, checks all axis-tilt magnitudes and the tool-rotation bound, and checks that the stored endpoint interval contains the generating c. The base-placement bound is not exercised in this case.

6.2.1 Independent tests of improved forward kinematics

The 16 forward/model cases in `test_kinematics_validation.py` use elementary sine/cosine matrices and explicit expected positions, rather than generating the answers with `ImprovedFwdKins`. For the tool-only case, use the nominal stage translation and wrist rotation

$$\boldsymbol{p}_c = [\theta_1, \theta_2, \theta_3]^T, \qquad R_w = R(\boldsymbol{e}_x, \theta_4)R(\boldsymbol{e}_y, \theta_5)R(\boldsymbol{e}_z, \theta_6).$$

Table 16: Independent forward/model checks.

Input	Expected answer or assertion
One constant joint distortion (6 cases)	Each stage offset adds to its own coordinate; each wrist offset produces the corresponding elementary rotation and rotates the nominal tool translation.
Stage 2 tilted by $\gamma = 0.015$ about $\boldsymbol{e}_z$, travel 80 mm	$R = I$, $\boldsymbol{p} = [600 - 80\sin\gamma,\ 80\cos\gamma,\ 0]^T$.
Wrist 4 tilt 0.012 rad about $\boldsymbol{e}_y$, travel 0.7 rad	$R = R(\boldsymbol{e}_y, 0.012)R(\boldsymbol{e}_x, 0.7)R(\boldsymbol{e}_y, -0.012)$; verify the rotated tool translation independently.
Nonzero wrist and tool corrections	Check the order $R_w R(\boldsymbol{\alpha}_{\text{we}})$ and $\boldsymbol{p}_c + \boldsymbol{p}_{\text{cw}} + R_w(\boldsymbol{p}_{\text{we}} + \boldsymbol{\epsilon}_{\text{we}})$.
Base correction and coordinate turn	Check right multiplication of the base correction and cancellation of $Q(c)Q(-c)$ against an independent tracker-frame pose.
Distortion function (6 cases)	A known linear function is recovered at every selected encoder position, between positions and beyond both endpoints; the fixed home value is preserved.

The pose examples compare rotation-matrix entries and position components directly with absolute tolerance 2×10^{-8}, rather than using the error-metric implementation as the oracle. The distortion checks use absolute tolerance 10^{-12}.

6.2.2 Tests of improved inverse kinematics and unusable data

Twelve inverse cases check both joint-range endpoints, an interior configuration, a configuration with $\theta_5 = \pi/2 - 10^{-5}$, selection of both feasible nominal wrist branches by the seed, the $\theta_6 = \pi$ endpoint with nonperiodic encoder distortion, an unreachable target, an iteration-limited solve, and four invalid targets (nonfinite position, scaled rotation, reflection and wrong type). Every successful inverse case recomputes the returned forward pose. Endpoint and near-singularity matrix/position comparisons use absolute tolerance 2×10^{-7}.

The iteration-limit case initially returned an unconverged command; nonorthogonal and reflected target rotations could also be accepted. Target type, finiteness and proper rotation are now validated before solving. The solver now checks the residual at the actual returned command and raises `ValueError` if no candidate satisfies the limits and tolerance. Already valid encoder angles are preserved; any wrapped candidate is checked again because encoder distortion need not be 2π-periodic.

Seven initialization cases reject a missing reference, a missing joint collection, only one non-reference stop, zero stage excitation, zero wrist excitation, parallel stage directions and a rank-deficient wrist-center system. The last four initially produced invalid directions or an undetermined center; explicit excitation/rank checks now reject them.

All 81 cases pass: 23 supplied simulator cases, 23 original student cases and 35 additional cases. Synthetic and numerical cases are deterministic; simulator-backed noisy tests retain their original tolerances. Passing these cases does not establish correctness at every configuration or under every allowed error realization.

6.3 Controlled sensitivity experiments

As in A1's frame-error propagation experiments, vary one experimental setting while holding the others fixed. The independent reference chain uses nonzero axis tilts, tool error and base placement, with smooth joint distortion

$$\Delta\theta^{\mathrm{rob}}_{k,\mathrm{sys}}(\theta_k) = \begin{cases} 0.04\sin(\theta_k/120), & k = 1,2,3 \quad \text{(mm)}, \\ 0.00004\sin(\theta_k/0.8), & k = 4,5,6 \quad \text{(rad)}. \end{cases}$$

The derivative magnitudes are bounded by $0.04/120 < 5 \times 10^{-4}$ for stages and $0.00004/0.8 = 5 \times 10^{-5}$ for wrist joints, consistent with the handout bounds. These functions are simulated test inputs, not an assumed form of the fitted distortion. They vanish at zero, so this experiment uses an exact home anchor. The fitted model still uses linearly interpolated distortion values.

For each independent arrival, add joint random error before evaluating the reference chain, then add a right-multiplied pose perturbation. Its independent rotation/translation components have standard deviations $s\sigma_R$ and $s\sigma_p$; joint standard deviations are $s(0.005\,\mathrm{mm})$ for stages and $s(10^{-4}\,\mathrm{rad})$ for wrist joints. Here s is the noise multiplier. This begins with pose observations and does not rerun marker registration or introduce tracker systematic distortion.

Table 17: Controlled experiment settings; other settings remain at baseline.

Setting varied	Values	Baseline
Independent arrivals per command	$1, 4, 16, 64$	16
Noise multiplier s	$0, 0.5, 1, 2$	1
Spacing multiplier	$0.5, 1, 2$	1

A spacing multiplier of one means 100 mm for stages and $\pi/4$ rad for wrist joints; both spacings change together. Each setting uses seeds 101–106 and the same 100 held-out full-range commands. The baseline is reused across comparisons. Fit in the fixed base-turn convention and measure tracker-frame forward RMSE, so uncertainty in the unobservable turn is not interpreted as an observable calibration error.

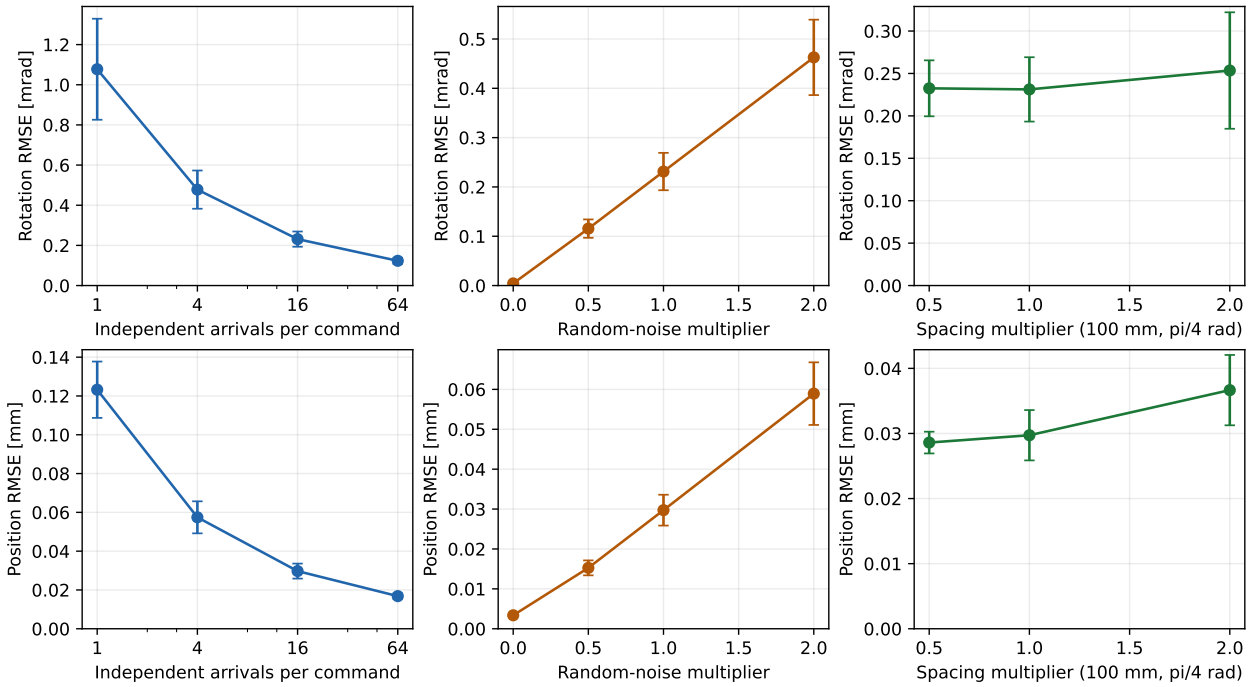


Figure 4: Controlled forward-error sensitivity. Points are mean RMSE over six seeds; bars are sample standard deviations across those seeds, not confidence intervals. The smooth reference distortion leaves interpolation error even at zero noise.

The driver `tests/kinematics_sensitivity.py` saves per-run and aggregate CSV files in `tests/data/`. Section 7.6 reports the numerical results. These experiments assess random-error and spacing sensitivity, not visibility, simulated travel time, homing uncertainty or tracker systematic bias.

6.4 Limitations

The end-to-end results cover one scene design and 25 seeds. The seed fixes the robot's hidden errors, its repeatability draws and the tracker's distortion field, but not the reading noise $\boldsymbol{\epsilon}_b$: the simulator draws it from NumPy's global random generator, which it does not seed. Repeated runs of one seed therefore give slightly different numbers. The original end-to-end tables come from the saved notebook run. Their tracker-frame results over 25 seeds stay within the targets (Section 7.4); the controlled experiments above are a separate data set. The template's grid covers the wrist joints to ± 1.3 and ± 2 rad rather than their full ranges; the full ranges are covered by the extended evaluation of Section 7.5. The rotation c about $\boldsymbol{d}_1$ cannot be observed, so poses relative to the robot base are known only to within its admissible interval. The original fitting tests reuse our model to generate data; the additional forward cases use independent analytic answers. Neither set proves accuracy for every hidden simulator model.

7 Results

Sections 7.1–7.5 preserve the original saved notebook run of the consecutive-command plan described in Section 4.3; the tracker's reading noise is not seeded, so a new run gives slightly different values (Section 6.4). Sections 7.1–7.3 report the main run (seed 1, single-axis plan). Section 7.4 repeats the pipeline over 25 seeds and compares the two plans, and Section 7.5 evaluates the handout's base-frame definition (notebook, extended evaluation).

7.1 Calibration data and time

Table 18: Calibration data, fit and simulated time (seed 1).

Quantity	Value
Homings H	5
Readings per stop, stage / wrist	7 / 68
Distinct poses / readings	39 / 1716
Fit parameters / Jacobian rank	59 / 59
Condition number of the weighted Jacobian	9.1×10^3
RMS residual, decoupled start	3.8×10^{-4} rad, 0.021 mm
RMS residual, after the fit (9 evaluations)	3.6×10^{-4} rad, 0.016 mm
Homing time	0.4 s
Diagnostic baseline (Step 6, not used by the fit)	83.8 s
Calibration data	543.3 s
Total calibration time	627.5 s
Testing time (40 nominal and 40 improved moves)	470.8 s

The collection time equals the planned estimate of 543.3 s. The decoupled start already leaves residuals close to the final ones, and the weakest Jacobian directions are individual stage distortion values. The saved original notebook run took about 4 minutes of computation, within the 10-minute limit. Additional tests and controlled sensitivity experiments are reported separately below.

7.2 Estimated parameters

Table 19 compares the estimates with the hidden ground truth (validation only). The estimated axes are within 0.0056° of the true angles between axes, and the stage axes are within 0.006° of their true directions in the tracker frame. The chosen rotation about $\boldsymbol{d}_1$ is $c = -0.0152$ rad, in the admissible interval $[-0.0179, -0.0125]$ rad.

Table 19: Estimated parameters and ground truth (seed 1; joints 1–3 in mm, 4–6 in rad).

	1	2	3	4	5	6
Joint offset, estimated	0.0210	0.0471	−0.0645	0.0074	0.0086	−0.0080
Joint offset c_k, true	0.0253	0.0697	−0.0583	0.0068	0.0080	−0.0083
	α_x	α_y	α_z	ϵ_x	ϵ_y	ϵ_z
Tool parameters, estimated	−0.0002	−0.0049	−0.0012	0.0577	−0.2026	−0.2301
Tool parameters, true	−0.0001	−0.0049	−0.0012	0.0581	−0.2091	−0.2358

The estimated joint offset is the value at $\theta_k = 0$ of the least-squares line through the knot values; the true value c_k is the constant term of the simulator's distortion, whose ripple the line does not represent, so differences of a few 0.01 mm are expected (here at most 0.023 mm).

7.3 Accuracy

Table 20: Forward and inverse errors in the tracker frame (seed 1). Targets: 10^{-3} rad and 0.2 mm.

	N	RMS [rad]	Max [rad]	RMS [mm]	Max [mm]	Met
FwdK, nominal	729	3.24×10^{-2}	5.17×10^{-2}	7.431	14.887	no
FwdK, improved	729	1.27×10^{-4}	2.06×10^{-4}	0.069	0.121	yes
InvK, nominal	40	3.47×10^{-2}	5.12×10^{-2}	7.691	13.088	no
InvK, improved	40	1.76×10^{-4}	3.97×10^{-4}	0.072	0.113	yes

Calibration reduces the maximum forward errors by a factor of about 250 in rotation and 120 in position. The improved maxima use 21% (forward) and 40% (inverse) of the angular target and 60% and 57% of the position target. The inverse errors include the repeatability of the final move.

7.4 Several seeds and the two plans

Table 21: Maximum errors in the tracker frame over five seeds (without the diagnostic baseline).

Plan	Seed	t_{cal} [s]	t_{test} [s]	FwdK [rad]	FwdK [mm]	InvK [rad]	InvK [mm]
single axis	1	528.8	435.0	1.78×10^{-4}	0.130	3.94×10^{-4}	0.116
single axis	2	528.8	435.0	2.34×10^{-4}	0.099	3.78×10^{-4}	0.094
single axis	3	528.8	435.0	2.33×10^{-4}	0.095	4.36×10^{-4}	0.109
single axis	4	528.8	435.0	2.03×10^{-4}	0.082	5.50×10^{-4}	0.092
single axis	5	528.8	435.0	1.35×10^{-4}	0.131	5.25×10^{-4}	0.128
concurrent	1	442.4	435.0	2.03×10^{-4}	0.104	5.14×10^{-4}	0.127
concurrent	2	442.4	435.0	2.29×10^{-4}	0.099	3.73×10^{-4}	0.099
concurrent	3	442.6	435.0	2.89×10^{-4}	0.110	3.89×10^{-4}	0.112
concurrent	4	442.6	435.0	1.74×10^{-4}	0.087	3.54×10^{-4}	0.104
concurrent	5	442.4	435.0	2.38×10^{-4}	0.092	4.65×10^{-4}	0.107

Both plans meet both targets for every seed. In 20 further runs of the single-axis plan (seeds 6–25) all targets are met as well; the worst errors are 2.9×10^{-4} rad and 0.146 mm forward and 5.2×10^{-4} rad and 0.142 mm inverse. The concurrent plan saves 86 s with similar accuracy. We keep the single-axis plan because its joint collections supply the decoupled initializer. Tracker systematic errors can still affect the estimates; these numerical results do not prove that base and tool parameters absorb them completely.

7.5 Handout's base-frame definition

The handout defines the forward error relative to the robot base, $(F^*_{\text{ee}}(\boldsymbol{\theta}))^{-1}\,\text{ImprovedFwdKins}(\boldsymbol{\theta})$. This error contains the error of the unobservable rotation c about $\boldsymbol{d}_1$ (Section 4.7), which no tracker data can reduce; only the handout's bounds limit it. Table 22 evaluates 600 random configurations over the full joint ranges for seeds 1–8: in the tracker frame, relative to the base with c chosen from

the bounds, and relative to the base with the fixed convention $\boldsymbol{\phi}_2 \cdot \boldsymbol{e}_x = 0$ ($c = 0$). The last pair of error columns gives the inverse error for 30 targets specified relative to the actual base.

Table 22: Maximum errors on random full-range configurations (seeds 1–8; angles in mrad, positions in mm). Targets: 1 mrad and 0.2 mm.

	FwdK						InvK		
	Tracker		Base, bounds		Base, $c = 0$		Base		Interval
Seed	mrad	mm	mrad	mm	mrad	mm	mrad	mm	of c [mrad]
1	0.23	0.143	0.33	0.110	15.6	11.64	0.45	0.089	5.3
2	0.21	0.127	0.87	0.556	3.4	2.45	1.00	0.485	6.1
3	0.24	0.089	6.97	4.863	14.8	10.50	7.07	3.926	15.7
4	0.33	0.135	1.88	1.129	14.9	10.41	1.89	0.978	4.5
5	0.26	0.109	0.57	0.312	11.8	8.75	0.63	0.237	4.6
6	0.35	0.134	1.39	0.790	17.3	12.74	1.44	0.698	5.3
7	0.23	0.125	3.93	2.713	9.2	6.46	3.98	2.474	10.1
8	0.26	0.118	2.59	1.736	10.0	7.29	2.62	1.334	14.0

In the tracker frame both targets are met over the full joint ranges. Relative to the base they are met for seed 1 only, for the forward and the inverse error alike. The unobservable turn contributes to the base-relative angular error and can produce millimetre-scale position error over a lever arm of about 700 mm. If the true turn lies between the accepted endpoints, choosing their midpoint limits its distance to that true turn to half their separation. This does not bound the complete pose error, which also contains axis, tool, distortion and measurement effects. Choosing c from the handout's bounds reduces the base-frame errors by a factor of 2–47 compared with the fixed convention, but it cannot meet the targets when the bounds leave a wide interval. We therefore report the measured base-relative errors and state that these targets are not met for all tested seeds.

7.6 Additional function tests and controlled experiments

All 81 automated test cases pass. The additional 35 cases verify independent forward examples, selected distortion values, inverse branch/limit/convergence behavior and rejection of unobservable initialization data (Section 6). The inverse-convergence, invalid-target and four unobservable-data cases exposed missing checks, which have been added.

Table 23 summarizes the separate controlled experiment of Section 6.3; Figure 4 includes all settings. Each number is mean tracker-frame forward RMSE over six seeds, evaluated on the same 100 held-out full-range commands. It is not a maximum-error compliance result.

Table 23: Selected controlled sensitivity results. Other settings remain at baseline.

Setting	Value	Rotation RMSE [mrad]	Position RMSE [mm]
Arrivals per command	1	1.077	0.1232
	64	0.123	0.0168
Noise multiplier	0	0.004	0.0034
	2	0.463	0.0589
Spacing multiplier	2	0.253	0.0367
	0.5	0.232	0.0286

Increasing arrivals from 1 to 64 reduces both errors. At zero random noise, the remaining RMSE is about 0.00431 mrad and 0.00337 mm: the smooth reference distortion is not exactly represented by the selected linear segments. Doubling both spacings increases the observed position RMSE. Halving the default spacing changes angular RMSE little at this noise level, so finer sampling is not uniformly better in these finite noisy data. These trends apply to the stated reference chain, pose-noise model and six seeds; they do not establish a universal accuracy guarantee.

The updated notebook was also executed end to end with the guarded inverse solver, taking about 115.1 s. All tracker-frame maximum-error checks in the single-axis seed 1–25 runs and concurrent seed 1–5 runs met both targets in this additional run. The base-relative evaluation recorded the failures below; errors among successful solves must not conceal these failures.

Table 24: Additional base-frame inverse check: 30 attempted targets per seed.

Seed	1	2	3	4	5	6	7	8
Failed solves	0	0	0	0	0	0	1	0

A failure means that neither refined branch passed the final residual and limit checks; it does not alone distinguish an unreachable model target from numerical failure. The complete check, including successful-solve errors and counts, is stored in `tests/data/kinematics_pipeline_check.json`. Global tracker reading noise remains unseeded, so these failure counts describe this run.

The original end-to-end tables above retain their saved data. The new per-run and aggregate CSV files are stored separately, and the experiment driver regenerates both the data and figure. This validation used Python 3.12, NumPy 2.5.4 and SciPy 1.18.1; it does not replace the original Python 3.11 run.

References

[1] R. H. Taylor, *Assignment 2 – CIS I (601.455/655), Fall 2026*, October 15, 2026, `2026_assignment_2.pdf`.

[2] LCSR-CIIS, *CIS I Course repository*. `https://github.com/LCSR-CIIS/CIS_I_Course`.